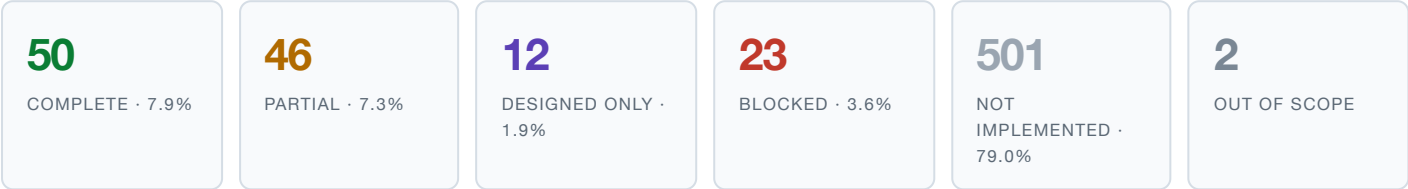
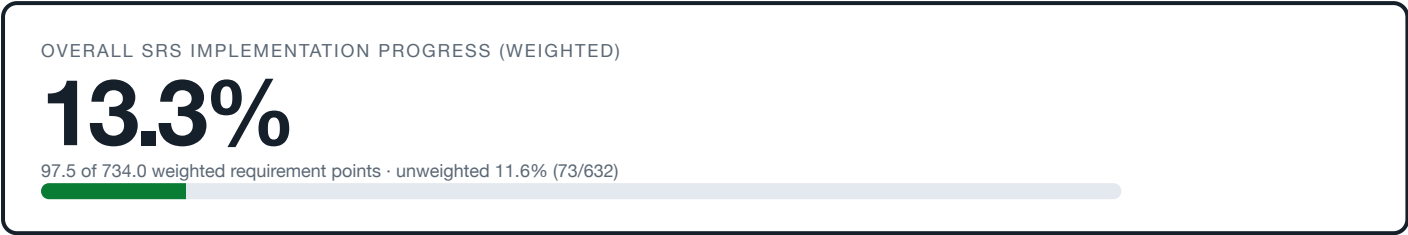


ROS — SRS Implementation Progress Audit

Restaurant Operating System backend · audited against **ROS_SRS_v1.0.pdf**
Audit date **2026-08-16** · Repository commit `48a16f92743d` · Branch `main`



1. Executive Summary

The repository contains a **rigorously built, well-verified foundation covering roughly one eighth of the SRS**. Six of the roughly twenty SRS domains have executable code: Identity/Auth, Tenancy, RBAC, Audit, Organisation, Catalogue and Inventory. Everything that generates revenue — Sales/POS, Payments, Kitchen Ops, Finance, Procurement, Workforce, CRM, Reporting, Offline — **does not exist in any form**.

What exists is unusually well engineered for its stage: PostgreSQL Row-Level Security is enabled and *forced* on all 43 tenant-scoped tables with a fail-closed predicate, verified live against the non-bypassing runtime role; tenant-safe composite foreign keys make cross-tenant edges structurally unrepresentable; the stock ledger is append-only at the privilege level; and the audit trail is SHA-256 hash-chained. 384 automated tests pass (136 unit, 248 E2E).

The honest headline is therefore two-sided. **Depth is high, breadth is very low**. The 13.3% weighted figure is not a criticism of the work done — it reflects that the project has deliberately built foundations first, gating each phase behind a ratified design. The critical risk is different: **23 requirements are BLOCKED** on domains that do not exist, and the single largest domain (Sales/POS, 76 FRs) is also the one every remaining financial, costing and reporting requirement depends on.

Most material finding. Recipe/Production Spec (FR-MNU-040–050, BR-MNU-001–003) has a **CLOSED/RATIFIED design gate** and is frequently spoken of as done. It has **zero executable code**: no `production` schema, no Prisma model, no migration, no service, no route, no test. It is scored **DESIGNED / AUTHORIZED — NOT IMPLEMENTED** at 0%. Design documents are not implementation.

Four SRS requirement IDs are referenced but never defined in ROS_SRS_v1.0.pdf: `FR-INT-020` , `FR-PLT-041` , `FR-RPT-055` , `FR-SEC-018` . They appear only in cross-references and traceability tables. They are excluded from the 634-requirement inventory and should be resolved in the SRS.

2. Overall Progress Dashboard

Status	Count	Share	Distribution
COMPLETE	50	7.9%	
PARTIALLY IMPLEMENTED	46	7.3%	
DESIGNED / AUTHORIZED - NOT IMPLEMENTED	12	1.9%	
BLOCKED	23	3.6%	
NOT IMPLEMENTED	501	79.0%	
OUT OF SCOPE	2	0.3%	
Total defined requirements	634	100%	

Metric	Value	Basis
Weighted implementation progress	13.3%	97.5 / 734.0 points. FR=1, BR=1, NFR=1.5, critical security=2, critical architectural=2
Unweighted requirement completion	11.6%	73 / 632 requirements, credit-weighted

Requirements fully COMPLETE	7.9%	50 of 634
Requirements with any executable code	15.1%	COMPLETE + PARTIAL = 96
Domains with any implementation	7 of 29	Identity, Tenancy, RBAC, Audit, Organisation, Catalogue, Inventory

3. Methodology

Progress was measured **against the SRS requirement inventory**, never against file counts, commit counts, migrations, endpoints or lines of code.

- 1. **Inventory.** `R0S_SRS_v1.0.pdf` was extracted with `pdftotext -layout` and parsed for every `FR-*`, `BR-*` and `NFR-*` identifier, capturing ID, MoSCoW priority, section and full text. **634 defined requirements** were recovered from 638 unique identifiers; the 4-identifier delta is the dangling-reference finding above.
- 2. **Repository audit.** All 221 TypeScript source files, 52 Prisma models, 13 migrations, 114 HTTP routes, 24 unit suites and 16 E2E suites were enumerated. All 8 ADRs and every phase/design document were read.
- 3. **Live database verification.** Assertions about RLS, grants, constraints, partitioning and tenant isolation were confirmed by querying the running PostgreSQL 16 instance as the restricted runtime role `ros_app`, with positive controls beside every negative assertion.
- 4. **Status assignment.** Each requirement received an explicit, individually justified status. Domains with no code received a per-domain justification recorded against every requirement in that domain.
- 5. **Scoring.** COMPLETE=100%, PARTIAL=50%, DESIGNED=0%, NOT IMPLEMENTED=0%, BLOCKED=0%. OUT OF SCOPE and NOT APPLICABLE are excluded from the denominator.

Deliberate strictness. A requirement was never marked COMPLETE because a schema, DTO, migration, ADR or design document exists, nor because a build passes, nor because an earlier report claimed completion. Executable implementation plus verification evidence was required. Where evidence was ambiguous the requirement was marked PARTIAL and the ambiguity recorded.

Weighting rationale. The default weighting in the audit brief was used unchanged. "Critical security" was operationalised as every `FR-SEC-*`, plus `FR-AUD-001..010`, `FR-PLT-010..014` (RLS and isolation) and `FR-API-010..014` (token security). "Critical architectural" was operationalised as `FR-PLT-003`, `FR-PLT-025`, all `BR-CORE-*`, all `BR-INV-*`, `BR-POS-004` and `BR-MNU-002`. Weighting the security requirements at 2 lowers the headline number, because most of them are unimplemented.

4. Domain Progress

Domain	Total	Complete	Partial	Designed	Blocked	Not impl.	Progress	
Inventory	53	24	9	0	13	7	55%	
Platform & Tenancy	22	7	3	0	0	12	45%	
Audit	10	3	2	0	0	5	40%	
API Platform	14	2	5	0	0	7	36%	
Catalogue / Menu / Recipe	42	6	9	12	0	13	26%	
Core Money & Units	4	1	0	0	0	3	25%	
NFR - Data	2	0	1	0	0	1	25%	
Security & Authorization	45	4	10	0	0	31	20%	
Data Retention & Migrations	10	1	1	0	0	8	15%	
Branch Ops & Central Kitchen	30	2	3	0	10	15	12%	
NFR - Observability	7	0	1	0	0	6	7%	
NFR - Performance	16	0	1	0	0	15	3%	
Localisation & Country Packs	20	0	1	0	0	19	2%	
Sales / POS	80	0	0	0	0	80	0%	
Offline & Sync	36	0	0	0	0	36	0%	
Procurement	33	0	0	0	0	33	0%	
Finance & Treasury	32	0	0	0	0	32	0%	
Costing & Analytics	31	0	0	0	0	31	0%	
Workforce	29	0	0	0	0	29	0%	
CRM & Loyalty	26	0	0	0	0	26	0%	
Kitchen Ops (KDS)	23	0	0	0	0	23	0%	

Reporting	18	0	0	0	0	18	0%	
DevOps & Release	15	0	0	0	0	15	0%	
NFR - Usability	11	0	0	0	0	11	0%	
NFR - Reliability	7	0	0	0	0	7	0%	
Integrations	6	0	0	0	0	6	0%	
NFR - Portability	6	0	0	0	0	6	0%	
Quality Assurance	5	0	0	0	0	5	0%	
NFR - Capacity	1	0	0	0	0	1	0%	

5. What We Have Actually Finished

Answering: if development stopped today, what functionality do we genuinely have? Judged strictly.

5.1 Production-ready and verified

Capability	Evidence
Tenant isolation via PostgreSQL RLS	ENABLE + FORCE on all 43 tenant tables, 4 policies each, fail-closed <code>NULLIF(current_setting('app.tenant_id',true))</code> predicate. Verified live as <code>ros_app</code> (NOBYPASSRLS): 0 rows without context, 0 cross-tenant, positive controls pass. 3 dedicated RLS E2E suites.
Tenant-safe composite foreign keys	ADR 0008 D-09. Every cross-aggregate edge references <code>(tenant_id, id)</code> , making cross-tenant parenting structurally unrepresentable. Verified in <code>pg_constraint</code> and by rejection tests.
Authentication & session lifecycle	Email/password, 15-minute access tokens, 30-day refresh with rotation and reuse detection revoking the whole family. ADR 0001/0005. <code>auth</code> , <code>refresh</code> , <code>password</code> E2E suites.
RBAC (tenant-scoped)	Roles, permission catalogue, role-permission and membership-role assignment, guard-enforced on all 114 routes. <code>rbac.e2e-spec.ts</code> .
Tamper-evident audit trail	SHA-256 hash chain per tenant, append-only enforced by privilege (REVOKE UPDATE/DELETE/TRUNCATE), verified live. ADR 0007.
Organisation hierarchy	Tenants, brands, branches, warehouses, central kitchens, stations, tables, operating hours, print routing, plus the <code>org.locations</code> registry. ADR 0008, 17 decisions ratified.
Append-only stock ledger	<code>inventory.stock_movements</code> RANGE-partitioned over 14 monthly partitions, INSERT+SELECT only, RLS on parent and every partition. BR-INV-001 verified live.
Inventory core	Stock items, batches, levels, movements, transfers with counterpart pairing, count sessions, waste, reason codes, FIFO/FEFO selection and weighted-average/standard/FIFO valuation. 74 tests.
Catalogue core	Menus, categories, items, variants, modifier groups, price lists, availability rules with 86-ing, branch menu resolution by priority with ambiguity warning.

5.2 Implemented but needing more verification

- **Audit coverage (FR-AUD-001).** `AuditService.record()` is called across identity, organisation, catalogue and inventory, but there is no gate proving every state-changing operation emits an entry.
- **NFR-PERF-005 (3,000-item stock query < 500 ms p95).** The supporting index exists; no measurement has ever been taken. The Inventory design gate itself says "to be measured, not assumed".
- **OpenAPI (NFR-API-001).** Swagger is wired at `/docs`, but decorator coverage is thin and the output is OpenAPI 3.0, not the required 3.1.
- **FR-SEC-048 parameterised queries.** Prisma parameterises by default and the single `$queryRawUnsafe` in `reconciliation.service.ts` uses bound parameters — but no static-analysis gate enforces the prohibition, so the guarantee is convention, not control.
- **BR-INV-002 transfer pairing.** Enforced in the service and E2E tested; no database-level guarantee that every `transfer_out` has exactly one `transfer_in`.

5.3 Design-only (authorized, zero code)

- **Production Spec / Recipes.** `docs/production/PRODUCTION_SPEC_DESIGN_GATE.md` is CLOSED/RATIFIED with D-17-02...D-17-08, GAP-1 and GAP-2 all resolved. Five tables designed, none created. Covers FR-MNU-040–050 and BR-MNU-001–003.

5.4 Not started

Sales/POS (76 FRs) · Offline & Sync (36) · Procurement (33) · Costing & Analytics (31) · Workforce (29) · Finance & Treasury (27) · CRM & Loyalty (26) · Kitchen Ops (23) · Reporting (18) · DevOps & Release (15) · Integrations (6) · Quality Assurance (5) · plus the great majority of Localisation, NFR-Performance, NFR-Usability, NFR-Reliability, NFR-Observability and NFR-Portability.

6. What We Thought Was Done But Isn't — Potential False Positives

Discrepancies between documentation claims, phase-completion claims, prior implementation reports, tests and actual source code.

Claim	Scored as	Why it must not count as complete
"Production Spec is ratified and authorized"	DESIGNED, 0%	The design gate is genuinely closed, but there is no production schema, model, migration, service, controller, route or test . 11 FRs and 3 BRs sit at 0%. This is the single largest overcounting risk in the project.
"Phase 15 Organisation complete"	Mostly COMPLETE	Complete for the entities it defines, but ADR 0008 deferred D-02 branch-scoped RBAC, D-10 branch groups and D-11 org.settings . Those deferrals leave FR-SEC-002/003/004, FR-BRN-005 and the entire FR-PLT-025–028 settings-resolver block unimplemented.
"Phase 16 Catalogue complete"	PARTIAL	13 tables and 40 routes exist, but price resolution is not implemented: <code>valid_from</code> , <code>valid_to</code> , <code>recurrence_rule</code> and <code>priority</code> are stored and returned, never evaluated. FR-MNU-021/022/023 are not satisfied by storing their columns.
"Inventory complete, 248 E2E passing"	PARTIAL	Verified accurate for what it covers — but only 22 of 50 FR-INV are COMPLETE . 13 are BLOCKED on Procurement, Sales, Treasury, Governance, Workforce and Sync; 5 have logic but no scheduler to deliver alerts; 7 were dropped by explicit decision.
"RLS is fully enforced"	COMPLETE (now)	True today, but it was false when the Inventory phase was first reported complete : the 14 ledger partitions had no RLS of their own, leaking every tenant's ledger to any direct-partition query. Caught only by a live probe, not by the test suite. Fixed in migration <code>20260817090000</code> . This is direct evidence that green tests did not equal a secure boundary.
"Audit trail satisfies FR-AUD"	3 of 10	Hash chain and append-only storage are excellent, but there is no chain-verification job (FR-AUD-005), no query, search, filter or export surface (FR-AUD-007/008), no retention (009) and no impersonation capture (010).
"Rate limiting implemented"	PARTIAL	ADR 0006 delivers IP-based throttling on <i>authentication endpoints</i> . FR-PLT-015 requires per-tenant limits on API requests, report generation and export jobs; FR-SEC-046 additionally requires progressive lockout . Neither exists.
"Multi-tenant SaaS platform"	0 of 4	Tenant <i>isolation</i> is excellent. Tenant <i>lifecycle</i> is absent: no self-service signup (FR-PLT-020), no read-only downgrade (021), no data export (022), no two-step termination (023).
Passing build / green suite	—	384 passing tests prove the implemented slice works. They cannot prove absent domains exist. Test count is not requirement coverage: 501 requirements have no code to test.

7. API Audit

114 routes implemented. The SRS §26.3 "Representative Endpoints" list is the requirement baseline; it is explicitly representative, not exhaustive.

Endpoint	SRS requirement	Exists	Functional	Auth	Valid.	Tenant iso.	Tests	Status
POST /auth/login	FR-SEC-020, FR-API-010	Yes	Yes	n/a (public)	Yes	Yes (RLS)	auth.e2e	COMPLETE
POST /auth/refresh	FR-API-010, FR-API-013	Yes	Yes	Token	Yes	Yes	refresh.e2e	COMPLETE
POST /auth/tenant	FR-PLT-003	Yes	Yes	JWT	Yes	Yes	tenant-context.e2e	COMPLETE
POST /auth/roles, /roles/:id/permissions	FR-SEC-001, FR-SEC-011	Yes	Yes	role.create/update	Yes	Yes	rbac.e2e	COMPLETE
POST /auth/terminals*	FR-SEC-028	Yes	Partial	terminal.manage	Yes	Yes	terminal.e2e	PARTIAL — no credential wipe
POST/GET /org/* (25 routes)	FR-PLT-001/002, FR-BRN-001/002	Yes	Yes	org perms	Yes	Yes	organisation.e2e	COMPLETE
POST/GET /catalogue/* (40 routes)	FR-MNU-001..030	Yes	Partial	6 catalogue perms	Yes	Yes	catalogue.e2e	PARTIAL — no price resolution
GET /catalogue/branches/:id/menus	FR-MNU-003	Yes	Yes	item.read	Yes	Yes	menu-resolution.spec	COMPLETE
GET /inventory/levels	FR-INV-010	Yes	Yes	inventory.view	Yes	Yes	inventory.e2e	COMPLETE
GET /inventory/items/:id/movements	FR-INV-030	Yes	Yes	inventory.cost.view	Yes	Yes	inventory.e2e	COMPLETE
POST /inventory/transfers, /receive	FR-INV-031, BR-INV-002	Yes	Yes	transfer.create/receive	Yes	Yes	inventory.e2e	COMPLETE
POST /inventory/counts, /lines, /post	FR-INV-040..045	Yes	Yes	count.perform/post	Yes	Yes	inventory.e2e	COMPLETE
POST /inventory/waste	FR-INV-050	Yes	Yes	waste.record	Yes	Yes	inventory.e2e	COMPLETE
GET /health	NFR-OBS-007	Yes	Trivial	Public	n/a	n/a	app.e2e	PARTIAL — static payload only
GET /docs (OpenAPI)	NFR-API-001	Yes	Partial	Public	n/a	n/a	None	PARTIAL — OpenAPI 3.0, thin coverage
GET /v1/menu (effective menu)	SRS §26.3	No	—	—	—	—	—	NOT IMPLEMENTED
POST /v1/recipes + /versions + /publish	SRS §26.3, GAP-1	No	—	—	—	—	—	DESIGNED ONLY
/v1/orders/* (11 endpoints)	SRS §26.3, FR-POS-*	No	—	—	—	—	—	NOT IMPLEMENTED
/v1/purchase-orders, /goods-receipts, /supplier-invoices	SRS §26.3, FR-PRC-*	No	—	—	—	—	—	NOT IMPLEMENTED
/v1/sync/batch, /changes, /status	SRS §26.3, FR-OFF-*	No	—	—	—	—	—	NOT IMPLEMENTED

Cross-cutting API findings

Aspect	Status	Finding
Server-side authorization (FR-SEC-045)	COMPLETE	All 114 routes pass through <code>JwtAuthGuard</code> ; every business route additionally carries <code>@RequirePermission</code> . Verified by enumeration.
Input validation (FR-SEC-047)	COMPLETE	Global <code>ValidationPipe</code> with <code>whitelist</code> + <code>forbidNonWhitelisted</code> + <code>transform</code> ; unknown-field rejection E2E tested.
Error semantics (§26.2)	COMPLETE	404 (never 403) on cross-tenant access is implemented and tested in every phase, preventing existence disclosure.
Stable error codes (FR-API-001)	NOT IMPL.	Responses use the NestJS default shape. No stable machine-readable code field.
Localised errors (FR-API-002)	NOT IMPL.	No <code>Accept-Language</code> handling anywhere.
Idempotency (FR-API-020..023)	NOT IMPL.	No <code>Idempotency-Key</code> handling, no key store, no replay or 409-on-fingerprint-mismatch. Mandatory for financial endpoints.
Branch set in token (FR-API-012)	PARTIAL	Token carries <code>sub</code> , <code>tid</code> , <code>mid</code> . No scope set and no permitted branch set.
Pagination / filtering	PARTIAL	List endpoints return unbounded collections; the movement ledger is hard-capped at 200 rows. No cursor or page parameters anywhere.
Machine clients (FR-API-011/014)	NOT IMPL.	No OAuth client credentials and no API keys.

8. Database Audit

SRS-required object	Exists	Finding
Schema separation (§25.1)	6 of 14	Present: identity, governance, org, catalogue, inventory, kitchen. Absent: production, sales, ck, procurement, workforce, fiscal, treasury, crm, analytics, sync.
tenant_id on tenant-scoped tables (FR-PLT-003)	Yes	NOT NULL on all 43 tenant-scoped tables. Immutability enforced by RLS WITH CHECK plus composite FKs.
RLS enabled + FORCE (FR-PLT-010)	Yes	All 43 tables, 4 policies each. Predicate identical everywhere; fail-closed verified live.
Non-bypassing runtime role (FR-PLT-011)	Yes	ros_app NOSUPERUSER/NOBYPASSRLS; DDL only via ros_migrator.
Composite tenant-safe FKs (ADR 0008 D-09)	Yes	UNIQUE (tenant_id, id) on every tenant-scoped parent; all cross-aggregate children reference the composite.
Append-only ledger (BR-INV-001, §25.2)	Yes	GRANT SELECT,INSERT + REVOKE UPDATE,DELETE,TRUNCATE on parent and all 14 partitions; no update/delete policy. Verified live.
Append-only audit (FR-AUD-003)	Yes	Same pattern on governance.audit_entries, plus SHA-256 hash chain.
Time partitioning (FR-DR-001)	Partial	stock_movements RANGE-partitioned by occurred_at, 14 monthly partitions, no DEFAULT partition by design. Orders and audit entries — the other high-volume tables named by the SRS — are not partitioned (orders do not exist).
Automatic partition creation (FR-DR-002)	No	Explicitly deferred. Partition creation is manual; the earliest partition boundary is a live operational risk.
Forward-only versioned migrations (FR-DR-010)	Yes	13 migrations in source control; prisma migrate status clean; zero drift confirmed.
CHECK constraints	Yes	XOR location targeting, non-zero movement quantity, batch/reason conditionals, non-negative batch quantity, expiry ≥ production date, count scope, standard-cost presence.
Money as BIGINT minor units	Yes	Consistent across catalogue and inventory. But no Money value type exists in application code (BR-CORE-001/002 unimplemented).
6-dp quantities (BR-CORE-003)	Yes	NUMERIC(18,6) on every quantity column.
Archival to object storage (FR-DR-003)	No	Partitions retained indefinitely; no export path.
Per-tenant restore (FR-DR-021)	No	No logical per-tenant backup or restore capability.

Database layer is the strongest part of the system. 52 models, 13 migrations, zero drift, and structural enforcement (RLS + composite FKs + privilege revocation + CHECK constraints) rather than application-only validation. The gap is breadth — 8 of 14 required schemas do not exist — not quality.

9. Security Audit

Control	State	Evidence / gap
Tenant isolation (RLS)	Implemented + tested	3 dedicated RLS E2E suites, live <code>ros_app</code> probes with positive controls, composite FK rejection tests.
Authentication	Implemented + tested	Argon2 hashing, JWT access/refresh, rotation with reuse detection and family revocation.
Password lifecycle	Implemented, partly verified	Change/reset with session revocation and enumeration-safe responses (ADR 0005). No per-tenant policy, no breached-password check (FR-SEC-025).
RBAC	Implemented, tenant scope only	FR-SEC-002 scoped assignments not implemented. <code>membership_roles.branch_id</code> exists but is never read. FR-SEC-003/004/005 consequently partial or absent.
Server-side authorization	Implemented + tested	All 114 routes guarded; 401/403/404 semantics tested per phase.
Input validation	Implemented + tested	Global whitelist + forbidNonWhitelisted; UUID pattern validation rather than <code>@IsUUID()</code> for ULID-as-UUID ids.
Audit trail	Implemented, no surface	Hash-chained and append-only, but unqueryable, unverified by any job, unretained, no impersonation capture.
Rate limiting	Implemented (auth only)	IP-based on auth endpoints. No per-tenant limits, no progressive lockout.
Security headers	Implemented	<code>helmet</code> wired in <code>main.ts</code> ; CSP deliberately relaxed for the Swagger UI — worth revisiting before production.
Terminal / device identity	Implemented, partial	Registration, fingerprints, status transitions (ADR 0004). Pairing/activation deferred; no credential wipe on revocation (FR-SEC-028).
PIN authentication	Missing	FR-SEC-021/022 entirely absent — and PIN is the primary POS authentication method.
MFA	Missing	FR-SEC-023/024. MFA is mandatory by FR-SEC-024 for roles holding <code>security.user.manage</code> , <code>settings.tenant.manage</code> or <code>api.key.manage</code> .
Segregation of duties	Missing	FR-SEC-015/016/017. FR-SEC-016 requires blocking , not warning, on self-approval combinations. No mechanism exists.
Approval workflow	Missing	FR-SEC-030–035. Inventory exposes a <code>requires_approval</code> gate that <i>refuses</i> when approval is required, but nothing can ever grant it. 5 FR-INV requirements are blocked behind this.
Encryption at rest / KMS	Missing	FR-SEC-041/042/043. No envelope encryption, no per-tenant data keys, no KMS.
Secrets management	Missing	FR-SEC-050. Configuration via <code>.env</code> ; no runtime secret injection.
Dependency scanning	Missing	FR-SEC-049. No CI pipeline exists at all (<code>.github</code> absent), which also fails FR-PLT-014.
IP allow-listing / SIEM	Missing	FR-SEC-052/053.
Data protection & privacy	Missing	FR-SEC-060–064: no classification, retention job, data-subject requests, residency or backup verification.

Security posture summary. The controls that exist are of high quality and genuinely verified — isolation, authentication, authorization enforcement, validation and audit integrity. But **only 4 of 45 FR-SEC requirements are COMPLETE**. The absent controls are not peripheral: PIN auth, MFA, segregation of duties, approvals, encryption at rest, secret management and CI scanning are all mandatory [M] requirements, and several are prerequisites for the POS domain.

10. Non-Functional Requirements Audit

NFR group	Count	Status	Evidence & gap
NFR-PERF (16)	16	1 partial	No performance test harness exists. NFR-PERF-005 has a supporting index but no measurement. NFR-PERF-001/002/003/004/021 are client-side. 030/031/032 (API latency) have never been measured.
NFR-USA (11)	11	0	All are UI/interaction requirements. No client surface exists in this repository.
NFR-REL (7)	7	0	NFR-REL-011 (at-most-once financial effect via idempotency keys) is the critical one and is unimplemented — it pairs with FR-API-020–023.
NFR-OBS (7)	7	1 partial	<code>audit_entries.correlation_id</code> exists. No structured JSON logging, no tracing, no RED metrics, no business metrics, no PII redaction layer, no alerting, no per-tenant health view.
NFR-PORT (6)	6	0	Client platform and deployment portability. Only a <code>docker-compose.yml</code> for local Postgres exists.
NFR-API (2)	2	1 partial	Swagger wired but OpenAPI 3.0 with thin coverage; no deprecation policy.
NFR-DATA (2)	2	1 partial	7-year movement retention structurally preserved (partitions never dropped, no DELETE grant). 15-minute reporting freshness not applicable — no reporting.

NFR-CAP (1)	1	0	Local store capacity — no client.
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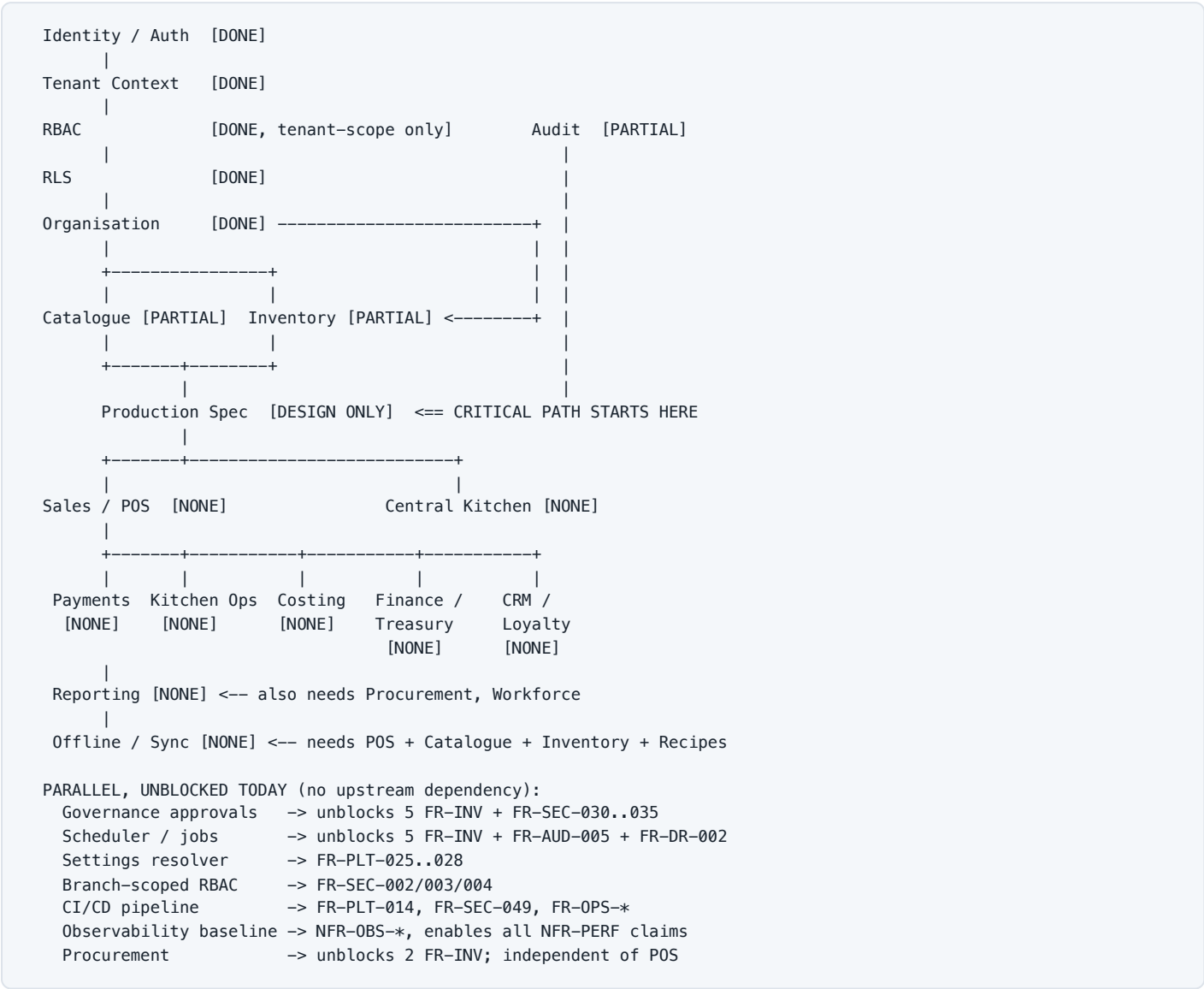
NFRs are the weakest area of the entire audit. 48 of 52 NFRs have no implementation and, more importantly, **no measurement capability**. Because NFRs carry weight 1.5, they depress the weighted score by design — correctly, since an unmeasurable NFR cannot be claimed. Establishing an observability and performance-measurement baseline is a prerequisite for ever asserting NFR compliance.

11. Phase & Design-Gate Audit

Phase	Design	Impl.	Tests	Prod-ready	SRS covered / still missing
Identity & tenancy (ADR 0001)	Complete	Complete	Yes	Near	FR-API-010/013, FR-SEC-001/011. Missing: MFA, PIN, API keys, tenant lifecycle.
Tenant context (ADR 0002)	Complete	Complete	Yes	Yes	FR-PLT-003/012. Branch scope explicitly deferred.
RLS (ADR 0003)	Complete	Complete	Yes	Yes	FR-PLT-010/011/012. Missing: FR-PLT-013/014 CI gates.
Terminal identity (ADR 0004)	Complete	Partial	Yes	No	FR-SEC-028 partial. Pairing/activation and credential wipe deferred.
Password lifecycle (ADR 0005)	Complete	Complete	Yes	Near	Missing FR-SEC-025 per-tenant policy and breached-password list.
Rate limiting (ADR 0006)	Complete	Partial	Yes	No	Auth endpoints only. FR-PLT-015 per-tenant and FR-SEC-046 lockout missing.
Audit trail (ADR 0007)	Complete	Partial	Yes	No	3 of 10 FR-AUD complete. No verification job, query surface or retention.
Organisation (ADR 0008, Phase 15)	Complete	Complete	Yes	Near	FR-PLT-001/002, FR-BRN-001/002. D-02, D-10, D-11 deferred → FR-SEC-002, FR-BRN-005, FR-PLT-025..028 missing.
Catalogue (Phase 16)	Complete	Partial	Yes	No	6 of 25 catalogue FRs complete. No price resolution, no auto-86, no bulk ops, no scheduling.
Inventory	Complete	Partial	Yes	Near	22 of 50 FR-INV complete, 13 blocked, 5 need a scheduler. One RLS bypass found and fixed post-"completion".
Production Spec	Complete	None	None	No	Gate CLOSED/RATIFIED with GAP-1 and GAP-2 resolved. Zero code. 11 FRs + 3 BRs at 0%.

Design completion vastly outruns implementation completion. Eleven phases have complete, ratified designs; seven have executable code; three are production-adjacent. The project's design discipline is a genuine asset — but design documents do not move the SRS progress number, and this audit does not credit them.

12. Dependency Graph & Critical Path



Critical path analysis

Finding	Detail
Current critical path	Production Spec → Sales/POS → Costing / Finance / Reporting. Production Spec is the only designed-and-authorized work item that unblocks nothing else until it is built; Sales/POS then unblocks the largest fan-out in the system.
Single biggest blocker	Sales/POS does not exist. 76 FRs directly, plus it is the upstream dependency for Costing (31), Finance (27), CRM (26), Kitchen Ops (23), Reporting (18), most of Offline (36) and 4 blocked FR-INV. Roughly 230 requirements sit downstream of it.
Highest-value next phase	Production Spec implementation. It is fully designed, fully ratified, has zero open questions, has all its prerequisites (Inventory UOM and stock items, Catalogue variants, Organisation scopes) in place, and is the declared next step. It converts 11 FRs and 3 BRs from 0% and unblocks Central Kitchen and recipe-linked depletion — the product's core differentiator per SRS §2.5.3.
Cheapest high-leverage work	Scheduler/jobs infrastructure. A single small capability converts 5 FR-INV from PARTIAL to COMPLETE (alerting/reconciliation delivery), enables FR-AUD-005 chain verification, and resolves FR-DR-002 automatic partition creation — which is a live operational risk, since the ledger has no DEFAULT partition and inserts past the last boundary will fail hard.
Can proceed in parallel	Governance approvals, scheduler, settings resolver, branch-scoped RBAC, CI/CD, observability and Procurement all have no upstream blocker and can be worked concurrently with Production Spec.

13. Remaining Work — Prioritized Backlog

Pri	Requirement IDs	Module	What must be implemented	Dependencies	Cplx	Why it matters
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P0	FR-MNU-040..045,049; BR-MNU-001,002	Production Spec	Implement the 5 ratified tables, migration, RLS, partial unique index, column-level grant, 4 endpoints, lifecycle, cycle detection, both test suites	Inventory (done), Catalogue (done), Organisation (done)	Medium	Only fully-designed, zero-question work item; unblocks Central Kitchen and recipe-linked depletion, the core product differentiator
P0	FR-DR-002; FR-AUD-005; FR-INV-011,024,051,066	Platform / Scheduler	Job scheduler: automatic partition creation 3 months ahead, audit chain verification, reorder/expiry/reconciliation alert delivery	None	Low	Ledger has no DEFAULT partition - inserts past 2027-09 will fail hard. Live operational risk today
P0	FR-PLT-014; FR-SEC-049; FR-OPS-003,004	Platform / CI	CI pipeline: RLS-coverage gate failing on any tenant_id table without forced RLS, dependency scanning, image signing, SBOM	None	Low	Two mandatory SRS gates currently have no enforcement whatsoever; the RLS partition bypass proves gates are needed
P0	FR-SEC-030..035; FR-INV-032,035,046,047,058	Governance	General approval mechanism: request entity, approver permission matching, sync and async decision, expiry, escalation, offline policy	RBAC (done)	Medium	Inventory already refuses when approval is required but nothing can grant it - 5 requirements are dead-ended
P1	FR-POS-* (76); BR-POS-001..004	Sales / POS	Order aggregate, lines, state machine, payments, tenders, discounts, comps, voids, refunds, snapshotting at sale	Production Spec, Catalogue pricing	Very high	Largest single domain; ~230 requirements sit downstream
P1	FR-MNU-020..023; FR-MNU-031..033	Catalogue	Price resolution engine (scope + validity window + recurrence + priority + order type) and auto-86 from stock	Inventory (done)	Medium	Price columns are stored but never evaluated - the feature is not delivered by its schema
P1	FR-SEC-002..005	Security / RBAC	Branch-scoped role assignments with validity dates; consume the existing membership_roles.branch_id	Organisation (done)	Medium	Mandatory; column already exists and is unused; blocks correct multi-branch authorization
P1	FR-SEC-021..024	Security / Auth	PIN authentication on registered terminals with lockout, and MFA (mandatory for privileged roles)	Terminal identity (partial)	Medium	PIN is the primary POS auth method; MFA is mandatory by FR-SEC-024
P1	FR-API-020..023; NFR-REL-011	API Platform	Idempotency-Key handling: key store, request fingerprint, 30-day retention, replay header, 409 on mismatch	None	Low	Mandatory on financially significant endpoints; prerequisite for safe POS and sync
P1	FR-AUD-005,007,008,009	Audit	Chain-verification job, audit query/search/filter/export surface, retention policy	Scheduler	Low	Audit data is captured but completely inaccessible - defeats its purpose
P1	FR-PLT-025..028	Platform	Hierarchical settings resolver with locking, inspector and effective dating	Organisation (done)	Medium	Deferred by ADR 0008 D-11; many downstream domains assume configurable settings
P1	NFR-OBS-001..007	Observability	Structured JSON logging with tenant/branch/correlation, tracing, RED metrics, PII redaction, alerting, per-tenant health	None	Medium	No NFR can be asserted without measurement; also required to operate the system at all
P2	FR-PRC-* (33)	Procurement	Suppliers, purchase orders, goods receipts, supplier invoices, three-way match	Inventory (done)	High	Unblocks FR-INV-005 and 067; independent of POS so can run in parallel
P2	FR-BRN-020..030	Central Kitchen	Production orders, distribution orders, yield variance, batch traceability	Production Spec	High	Directly enabled once Production Spec ships
P2	FR-PLT-020..023	Platform	Tenant lifecycle: self-service signup, read-only downgrade, data export, two-step termination	None	Medium	Mandatory for a commercial SaaS; entirely absent
P2	FR-KDS-* (23)	Kitchen Ops	Ticket display, routing, bump, timing	Sales/POS	High	Needs orders to exist
P2	FR-LOC-020..031	Localisation	Country pack entity, versioning, signing, conformance suite, tax engine registry	Settings resolver	High	All jurisdiction behaviour is meant to be pack-driven; nothing exists

P2	FR-SEC-041..043,050,060..064	Security / Infra	Encryption at rest, envelope encryption with per-tenant keys, KMS, secret manager, retention and DSR handling	Deployment platform	Medium	Mandatory, but infrastructure-layer rather than application-layer
P3	FR-OFF-* (36)	Offline / Sync	Local store, sync batch endpoints, HLC ordering, conflict register	POS, Catalogue, Inventory, Recipes	Very high	Depends on nearly everything upstream
P3	FR-RPT-*, FR-CST-*, FR-CRM-*, FR-HRM-*, FR-FIN-*	Multiple	Reporting, costing/analytics, CRM, workforce, finance	Sales/POS and Procurement	Very high	All require transactional data that does not yet exist
P3	NFR-PERF-*, NFR-USA-*, NFR-PORT-*	NFR	Performance harness and budgets; client-side usability and portability	Observability; client applications	Medium	Cannot be measured or claimed until subject systems exist

14. Recommended Next Steps

1. **Implement Production Spec now.** It is the only work item with a closed, ratified design, zero open questions and every prerequisite satisfied. Implementation authorization has already been issued.
2. **Ship a scheduler in the same cycle.** Small effort, disproportionate return: it closes FR-DR-002 (a live risk — the ledger has no DEFAULT partition and will reject inserts after the last boundary), FR-AUD-005, and converts 5 FR-INV from PARTIAL to COMPLETE.
3. **Stand up CI before adding more domains.** FR-PLT-014 and FR-SEC-049 are mandatory and unenforced. The partition RLS bypass — a real cross-tenant leak that green tests did not catch — is the argument for an automated RLS-coverage gate.
4. **Build the Governance approval mechanism.** Five Inventory requirements are already dead-ended behind an approval gate that nothing can satisfy, and FR-SEC-030–035 blocks discounts, refunds, purchase orders and price changes later.
5. **Then commit to Sales/POS as the next major domain.** Roughly 230 requirements sit downstream. Nothing in Costing, Finance, CRM, Reporting or Offline can start until orders exist.
6. **Establish observability before claiming any NFR.** 48 of 52 NFRs currently have no measurement capability. Until logging, tracing and metrics exist, no performance or reliability requirement can honestly be marked COMPLETE.
7. **Resolve the 4 dangling SRS identifiers** (FR-INT-020, FR-PLT-041, FR-RPT-055, FR-SEC-018) in the specification itself.

15. Final Conclusion

The ROS backend is **13.3% complete against the SRS on a weighted basis** (11.6% unweighted). Of 634 defined requirements, 50 are COMPLETE, 46 PARTIAL, 12 DESIGNED but unimplemented, 23 BLOCKED and 501 NOT IMPLEMENTED.

The work completed is of **high engineering quality and is genuinely verified** — structural database enforcement, forced RLS with live cross-tenant probes, append-only ledgers, hash-chained audit, 384 passing tests and zero schema drift. The phase-gate discipline, in which every design is ratified before implementation and blockers are raised rather than guessed, is a real asset and is visible throughout the repository.

The gap is **breadth, not depth**. Seven of roughly twenty-nine SRS domains have any code. Every revenue-generating and financial capability — ordering, payment, kitchen operations, procurement, workforce, finance, reporting and offline operation — is absent. Non-functional requirements are the weakest area, not because they were done badly but because **no measurement capability exists to claim them at all**.

The single most important correction this audit makes: a closed design gate is not implementation. Production Spec, Governance approvals, the settings resolver, branch-scoped RBAC and branch groups are all thoroughly designed and frequently discussed as settled — and all sit at **0% implemented**. Progress must be counted in executable, verified code.

Audit performed 2026-08-16 against commit `48a16f92743d5f8049743ab6f1f6022b50c0fac1`. No application code, schema, migration or test was modified. The only artifact generated is this report.

16. Complete Requirement Inventory – Requirement-by-Requirement Status

All 634 defined SRS requirements, grouped by domain. Priority is the SRS MoSCoW marker where present ([M]ust / [S]hould / [C]ould).

Inventory – 53 requirements · 55% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
BR-INV-001		StockMovement records SHALL be append-only. No UPDATE or DELETE is permitted. Reversals are new movements with opposite sign referencing the original.	COMPLETE	Evidence: Append-only ledger; REVOKE + no update/delete policy; verified live on parent and all 14 partitions
BR-INV-002		For every transfer_out there SHALL be exactly one transfer_in with equal absolute quantity, linked by counterpart_movement_id. In-transit stock is visible as the interval betw	PARTIAL	Evidence: transfers.service pairs transfer_out/transfer_in at insert; e2e tested · Missing: Service-enforced only; no database guarantee of pairing
BR-INV-003		The sum of all movements for an (item, location) pair SHALL equal the stock_levels projection for that pair. A reconciliation job SHALL verify this daily and raise an alert on	COMPLETE	Evidence: Projection written in the same transaction + on-demand reconciliation endpoint; tested
FR-INV-001	M	The System SHALL maintain a stock item master with the following attributes:	PARTIAL	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests · Missing: Catalogue linkage incomplete (C-04)
FR-INV-002	M	The base unit of measure SHALL be immutable once any stock movement exists for the item.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-003	M	The System SHALL support multiple purchase units per item with independent conversion factors, so an item may be bought in cases of 12 from one supplier and cases of 24 from a	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-004	M	The System SHALL support mass↔volume conversion for an item where a density factor is configured, and SHALL reject such conversion where it is not.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-005	S	The System SHALL support supplier-specific item codes and barcodes, so that a supplier's invoice or delivery note can be matched automatically.	BLOCKED	Missing: Needs Procurement supplier codes/barcodes · Depends on: Procurement
FR-INV-010	M	The System SHALL maintain current stock levels per (item, location) pair as a projection over the stock movement ledger.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-011	M	Stock levels SHALL be reconcilable to the movement ledger at any point in time, and a scheduled job SHALL verify the reconciliation daily and alert on divergence.	PARTIAL	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests · Missing: Logic implemented; scheduling/alert delivery deferred (C-08) - no scheduler exists
FR-INV-012	M	The System SHALL support three costing methods, configurable per item:	PARTIAL	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests · Missing: Standard-cost variance posting not implemented (no variance account entity in SRS)
FR-INV-013	M	Where an item is batch-tracked, FIFO valuation SHALL follow batch receipt order, and consumption SHALL record which batch was consumed.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-014	M	The System SHALL permit negative stock levels, recording them and raising an alert, rather than blocking the transaction that causes them.	PARTIAL	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests · Missing: Logic implemented; scheduling/alert delivery deferred (C-08) - no scheduler exists
FR-INV-015	M	The System SHALL report total inventory value per location, per category, and per item, at any historical date, computed from the movement ledger. 11.4 Batch and Expiry Manage	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-020	M	For batch-tracked items, the System SHALL create a batch record on receipt or production, carrying: batch number, production date, expiry date, quantity, unit cost, supplier,	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-021	M	The System SHALL default the expiry date from the item's configured shelf life, allowing override at receipt.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests

FR-INV-022	M	The System SHALL support two consumption strategies, configurable per item: • FIFO — consume the oldest received batch first. • FEFO — consume the batch with the nearest	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-023	M	FEFO SHALL be the default for items with expiry tracking enabled.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-024	M	The System SHALL generate expiry alerts at configurable horizons per item category (default: 7, 3, and 1 days before expiry).	PARTIAL	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests • Missing: Logic implemented; scheduling/alert delivery deferred (C-08) - no scheduler exists
FR-INV-025	M	The System SHALL provide an expiry action worklist showing items approaching expiry, with quantity, value, and one-tap actions: mark wasted, transfer to another location, or c	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-026	S	The System SHALL support automatic write-off of expired batches at day close, configurable per item category, creating waste records with reason expired.	BLOCKED	Missing: Needs Treasury day close • Depends on: Treasury
FR-INV-027	S	The System SHALL support batch traceability reporting: given a batch, list every order that consumed it (forward trace); given an order, list every batch consumed (backward tr	BLOCKED	Missing: Needs Sales for the forward trace half • Depends on: Sales
FR-INV-030	M	The System SHALL record every stock change as an immutable movement per the schema in 7.4.3.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-031	M	Inter-location transfers SHALL be modelled as a two-step process — dispatch and receipt — creating paired movements, with quantity in transit visible between them.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-032	M	The receiving location SHALL be able to record a received quantity differing from the dispatched quantity, creating a transfer discrepancy record requiring investigation and a	BLOCKED	Evidence: requires_approval gate exists (B-2) • Missing: Needs Governance approval workflow • Depends on: Governance
FR-INV-033	S	Transfers SHALL support a printed or digital transfer note with barcode/QR for scanning at receipt.	NOT Impl.	Missing: Unmet by explicit Inventory design decision (transfer note/QR, cycle counting, storage-ordered sheets, anomaly detection, forecasting)
FR-INV-034	M	Transfers between locations with different costing bases SHALL transfer at the sending location's cost, with the difference posted to a transfer variance account where configu	PARTIAL	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests • Missing: Cost transferred; transfer variance not posted
FR-INV-035	M	Manual stock adjustments SHALL require a reason code, SHALL be permission-gated, and SHALL require approval above a configurable value threshold.	BLOCKED	Evidence: requires_approval gate exists (B-2) • Missing: Needs Governance approval workflow • Depends on: Governance
FR-INV-040	M	The System SHALL support stock count sessions covering: a full location, a category, a storage area, or an ad-hoc item list.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-041	M	The System SHALL support two count modes: • Blind count — expected quantity is hidden during counting. • Open count — expected quantity is displayed.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-042	M	Blind count SHALL be the default and SHALL be enforceable as mandatory by policy configuration.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-043	M	The System SHALL support counting on a mobile device with barcode scanning, offline-capable, syncing on completion.	BLOCKED	Missing: Needs offline/sync for mobile counting • Depends on: Offline/Sync
FR-INV-044	M	The System SHALL freeze the expected quantity at the moment a count session opens, and SHALL account for movements occurring during the count window so that the variance calcu	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-045	M	On posting, the System SHALL compute per-item variance in quantity and value, and SHALL create count_adjustment movements bringing recorded stock to counted stock.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-046	M	Variances exceeding a configurable threshold (by percentage or value) SHALL require a recount or a written explanation before posting.	BLOCKED	Evidence: requires_approval gate exists (B-2) • Missing: Needs Governance approval workflow • Depends on: Governance
FR-INV-047	M	Count posting SHALL be permission-gated and SHALL be an approval- requiring action for high-value adjustments.	BLOCKED	Evidence: requires_approval gate exists (B-2) • Missing: Needs Governance approval workflow • Depends on: Governance

FR-INV-048	S	The System SHALL support cycle counting: a rolling schedule that counts high-value or high-variance items frequently and low-risk items rarely, so that a full count is never r	NOT IMPL.	Missing: Unmet by explicit Inventory design decision (transfer note/QR, cycle counting, storage-ordered sheets, anomaly detection, forecasting)
FR-INV-049	S	The System SHALL provide a count sheet ordered by physical storage location to minimise walking, configurable by the tenant.	NOT IMPL.	Missing: Unmet by explicit Inventory design decision (transfer note/QR, cycle counting, storage-ordered sheets, anomaly detection, forecasting)
FR-INV-050	M	Count sessions SHALL retain full history, including counted values, counter identity, timestamps, recounts, and approvals.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-051	M	A scheduled reconciliation job SHALL verify that the sum of movements equals the stock level projection for every (item, location) pair, and SHALL raise a platform alert on an	PARTIAL	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests · Missing: Logic implemented; scheduling/alert delivery deferred (C-08) - no scheduler exists
FR-INV-055	M	The System SHALL support waste recording at any point: from the POS (void disposition), from the kitchen (KDS), from a stock screen, and from a mobile device.	BLOCKED	Missing: Needs Sales and Kitchen Ops entry points · Depends on: Sales / Kitchen Ops
FR-INV-056	M	Every waste record SHALL capture: item, quantity, unit, reason code, location, employee, timestamp, and computed value. Photographs SHALL be optionally attachable.	BLOCKED	Missing: Needs Workforce employee identity and photo storage · Depends on: Workforce
FR-INV-057	M	The System SHALL provide a configurable reason code taxonomy, defaulting to:	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-058	M	Waste above a configurable value threshold SHALL require manager approval before posting.	BLOCKED	Evidence: requires_approval gate exists (B-2) · Missing: Needs Governance approval workflow · Depends on: Governance
FR-INV-059	S	The System SHALL distinguish “true waste” from “controlled consumption” (staff meals, tastings, marketing samples), reporting them separately so that the waste percentage is n	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-060	S	The System SHALL provide waste analysis by item, reason, employee, station, shift, day-part, and branch, with trend comparison.	BLOCKED	Missing: Needs Kitchen Ops / Workforce (station, shift, day-part) · Depends on: Kitchen Ops / Workforce
FR-INV-061	C	The System SHALL detect anomalous waste patterns using statistical baselines — for example, waste of a high-value item consistently recorded by one employee during a single sh	NOT IMPL.	Missing: Unmet by explicit Inventory design decision (transfer note/QR, cycle counting, storage-ordered sheets, anomaly detection, forecasting)
FR-INV-065	M	The System SHALL maintain per-location reorder points and reorder quantities per item.	COMPLETE	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests
FR-INV-066	M	The System SHALL generate low-stock alerts when available quantity falls below the reorder point.	PARTIAL	Evidence: src/modules/inventory/* (16 tables, 13 routes); inventory.e2e-spec.ts 36 tests; inventory-rls.e2e-spec.ts 18 tests; costing.spec.ts 20 tests · Missing: Logic implemented; scheduling/alert delivery deferred (C-08) - no scheduler exists
FR-INV-067	S	The System SHALL compute a suggested order quantity using: $\text{forecast_demand} = \text{average_daily_usage}(\text{last } N \text{ days, day-of-week weighted}) \times (\text{supplier_lead_time_days} + \text{revie}$	BLOCKED	Missing: Needs Procurement supplier lead time · Depends on: Procurement
FR-INV-068	S	The suggestion SHALL be capped by remaining shelf life: the System SHALL NOT suggest ordering more of a perishable item than can be consumed before expiry at forecast usage.	NOT IMPL.	Missing: Unmet by explicit Inventory design decision (transfer note/QR, cycle counting, storage-ordered sheets, anomaly detection, forecasting)
FR-INV-069	S	The System SHALL incorporate known future demand — scheduled events, catering orders, promotions — into the forecast where recorded.	NOT IMPL.	Missing: Unmet by explicit Inventory design decision (transfer note/QR, cycle counting, storage-ordered sheets, anomaly detection, forecasting)
FR-INV-070	C	The System SHALL adjust forecasts for known seasonality, including Ramadan, Eid, and local holidays defined in the country pack.	NOT IMPL.	Missing: Unmet by explicit Inventory design decision (transfer note/QR, cycle counting, storage-ordered sheets, anomaly detection, forecasting)

Platform & Tenancy — 22 requirements · 45% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
BR-PLT-001		Inventory is held at a location (branch, warehouse, or central kitchen), never at brand or tenant level. Two brands operating from the same physical kitchen must be modelled a	COMPLETE	Evidence: org.locations registry; stock held per location only; verified
FR-PLT-001	M	The System SHALL model tenants, brands, branches, warehouses, and central kitchens as distinct entities with the hierarchy above.	COMPLETE	Evidence: org.brands/branches/warehouses/central_kitchens + /org/* routes; organisation.e2e-spec.ts

FR-PLT-002	M	A tenant SHALL be able to own multiple brands, and a brand SHALL be able to own multiple branches.	COMPLETE	Evidence: org.brands/branches/warehouses/central_kitchens + /org/* routes; organisation.e2e-spec.ts
FR-PLT-003	M	Every tenant-scoped record SHALL carry an immutable tenant_id. Records SHALL NOT be transferable between tenants.	COMPLETE	Evidence: tenant_id NOT NULL on all 43 tenant tables; RLS FORCE; composite FKs (ADR 0008 D-09); rls.e2e-spec.ts
FR-PLT-004	S	A branch MAY be reassigned between brands within the same tenant by a Tenant Owner, with a full audit record and an explicit warning regarding menu and pricing implications.	PARTIAL	Evidence: POST /org/branches/:id/brand (ADR 0008 D-13) + audit · Missing: No explicit menu/pricing-implication warning surfaced
FR-PLT-010	M	The System SHALL enforce tenant data isolation at the database layer using PostgreSQL Row-Level Security, independent of application-layer filtering. Every tenant-scoped table	COMPLETE	Evidence: ENABLE+FORCE RLS, 4 policies/table, NULLIF(current_setting) predicate; verified live as ros_app
FR-PLT-011	M	The application database role SHALL NOT have BYPASSRLS. A separate, restricted migration/admin role SHALL be used for schema changes and platform-level operations.	COMPLETE	Evidence: ros_app NOSUPERUSER/NOBYPASSRLS; ros_migrator for DDL; verified in pg_roles
FR-PLT-012	M	Any request that reaches the data layer without a resolved tenant context SHALL fail closed with an error, never defaulting to an unfiltered query. Risk: The catastrophic fail	COMPLETE	Evidence: Fail-closed verified live: 0 rows on every tenant table with no app.tenant_id
FR-PLT-013	M	The CI pipeline SHALL execute a cross-tenant isolation test suite that, for every table containing tenant_id, attempts to read and write records belonging to Tenant B while th	PARTIAL	Evidence: rls/catalogue-rls/inventory-rls e2e suites cover 3 contexts · Missing: No CI pipeline; not every tenant_id table covered
FR-PLT-014	M	The CI pipeline SHALL fail if any table with a tenant_id column lacks an enabled and forced RLS policy.	NOT IMPL.	Missing: No CI pipeline exists (.github absent)
FR-PLT-015	M	The System SHALL apply per-tenant rate limits on API requests, report generation, and export jobs.	PARTIAL	Evidence: @nestjs/throttler on auth endpoints (ADR 0006) · Missing: IP-based not per-tenant; no limits on reports/exports
FR-PLT-016	M	Long-running analytical queries SHALL execute against a read replica using a separate connection pool, so that a heavy report cannot exhaust the transactional pool.	NOT IMPL.	Missing: No read replica, no statement timeouts, no isolation tier
FR-PLT-017	S	The System SHALL apply a statement timeout of 5 seconds on the transactional pool and 120 seconds on the analytical pool. 6.2.4 Enterprise Isolation Tier	NOT IMPL.	Missing: No read replica, no statement timeouts, no isolation tier
FR-PLT-018	C	For Enterprise tenants with contractual isolation requirements, the System SHALL support deployment into a dedicated schema or a dedicated database instance, using identical a	NOT IMPL.	Missing: No read replica, no statement timeouts, no isolation tier
FR-PLT-020	M	Tenant self-service signup SHALL create a working branch with a starter menu template, a default role set, and a first user in a single flow.	NOT IMPL.	Missing: No tenant lifecycle: signup flow, downgrade read-only, export, termination countdown
FR-PLT-021	M	The System SHALL NOT delete tenant data upon downgrade or suspension. Data beyond the plan's limits SHALL become read-only rather than being destroyed.	NOT IMPL.	Missing: No tenant lifecycle: signup flow, downgrade read-only, export, termination countdown
FR-PLT-022	M	At any lifecycle state except purged, a tenant SHALL be able to export all of their data in machine-readable format (CSV per entity plus a JSON manifest) within 24 hours of re	NOT IMPL.	Missing: No tenant lifecycle: signup flow, downgrade read-only, export, termination countdown
FR-PLT-023	M	Termination SHALL be a two-step process: an initiation that begins a 30- day reversible countdown, and a final confirmation. Data purge SHALL occur no earlier than 30 days aft	NOT IMPL.	Missing: No tenant lifecycle: signup flow, downgrade read-only, export, termination countdown
FR-PLT-025	M	The System SHALL implement a hierarchical settings resolver with the precedence above.	NOT IMPL.	Missing: Settings resolver absent; org.settings deferred by ADR 0008 D-11 · Depends on: ADR 0008 D-11
FR-PLT-026	M	A setting SHALL be markable as locked at any level, preventing override at lower levels. Locked settings SHALL be visibly indicated in the UI at lower levels with the locking	NOT IMPL.	Missing: Settings resolver absent; org.settings deferred by ADR 0008 D-11 · Depends on: ADR 0008 D-11
FR-PLT-027	S	The System SHALL provide a settings inspector showing, for any effective value, which level supplied it and what the value would be at each level.	NOT IMPL.	Missing: Settings resolver absent; org.settings deferred by ADR 0008 D-11 · Depends on: ADR 0008 D-11
FR-PLT-028	M	Settings that affect financial computation (tax class, rounding policy, service charge) SHALL be versioned with effective dates, and historical transactions SHALL be interpret	NOT IMPL.	Missing: Settings resolver absent; org.settings deferred by ADR 0008 D-11 · Depends on: ADR 0008 D-11

Audit — 10 requirements · 40% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-AUD-001	M	The System SHALL record an immutable audit entry for every state- changing operation.	PARTIAL	Evidence: AuditService.record() called from identity/org/catalogue/inventory services · Missing: Not proven for every state-changing operation; no coverage gate
FR-AUD-002	M	Each audit entry SHALL contain:	COMPLETE	Evidence: governance.audit_entries carries actor/entity/action/metadata/correlation; audit.e2e-spec.ts
FR-AUD-003	M	Audit entries SHALL be append-only. The application database role SHALL hold INSERT and SELECT grants on the audit table and SHALL NOT hold UPDATE or DELETE.	COMPLETE	Evidence: GRANT SELECT,INSERT + REVOKE UPDATE,DELETE,TRUNCATE; verified live (permission denied)
FR-AUD-004	M	Audit entries SHALL be hash-chained: each entry's hash covers its own content and the previous entry's hash, per tenant. hash(n) = SHA-256(canonical_json(entry_n) hash(n-1	COMPLETE	Evidence: SHA-256 hash chain per tenant (audit-hash.ts); unit + e2e tested
FR-AUD-005	M	A scheduled job SHALL verify chain integrity and SHALL raise a platform- level security alert on any break.	NOT IMPL.	Missing: No scheduler; no chain-verification job; no alerting
FR-AUD-006	M	The following actions SHALL always generate audit entries: authentication success and failure, permission changes, role changes, price changes, recipe changes, discounts, comp	PARTIAL	Evidence: auth success/failure, permission/role changes, price changes audited · Missing: discounts/comps/voids/refunds/cash variances impossible - domains absent
FR-AUD-007	M	Audit log access SHALL itself be audited.	NOT IMPL.	Missing: No audit query/search/filter/export API surface exists
FR-AUD-008	M	The audit log SHALL be searchable and filterable by actor, entity, action, date range, branch, and correlation ID, and SHALL be exportable by users with audit.view plus report	NOT IMPL.	Missing: No audit query/search/filter/export API surface exists
FR-AUD-009	M	Audit entries SHALL be retained for the greater of 7 years or the statutory period of the branch's jurisdiction.	NOT IMPL.	Missing: No retention policy or enforcement
FR-AUD-010	M	Support staff access to tenant data SHALL require an explicit impersonation session with a recorded reason, a time limit, tenant-visible notification, and full audit capture o	NOT IMPL.	Missing: No impersonation session mechanism

API Platform — 14 requirements · 36% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-API-001	M	Error responses SHALL use a stable machine-readable code in addition to the human-readable title and detail.	PARTIAL	Evidence: Nest default problem shape; 404-on-cross-tenant honoured and tested · Missing: No stable machine-readable error codes; no Accept-Language localisation
FR-API-002	M	Error messages SHALL be localised per the request's Accept-Language header.	PARTIAL	Evidence: Nest default problem shape; 404-on-cross-tenant honoured and tested · Missing: No stable machine-readable error codes; no Accept-Language localisation
FR-API-003	M	Error responses SHALL NOT leak internal details: stack traces, SQL, internal hostnames, or other tenants' data.	PARTIAL	Evidence: Nest default problem shape; 404-on-cross-tenant honoured and tested · Missing: No stable machine-readable error codes; no Accept-Language localisation
FR-API-010	M	Interactive clients SHALL authenticate using short-lived access tokens (15 minutes) with refresh tokens (30 days, rotating on use).	COMPLETE	Evidence: 15m access / 30d refresh with rotation; refresh.e2e-spec.ts
FR-API-011	M	Machine clients SHALL authenticate using OAuth 2.0 client credentials or scoped API keys.	NOT IMPL.	Missing: No OAuth client credentials, no API keys
FR-API-012	M	Tokens SHALL carry: subject, tenant, scope set, and permitted branch set. Every request SHALL be authorised against both the permission and the scope.	PARTIAL	Evidence: Access token carries sub, tid, mid · Missing: No scope set and no permitted branch set in token
FR-API-013	M	Refresh token rotation SHALL detect reuse of a consumed refresh token and SHALL revoke the entire token family on detection.	COMPLETE	Evidence: Refresh reuse detection revokes the token family; ADR 0001; tested
FR-API-014	M	API keys SHALL be displayed exactly once at creation, stored as a hash, and identifiable thereafter only by a non-secret prefix.	NOT IMPL.	Missing: No OAuth client credentials, no API keys
FR-API-020	M	Every POST and PATCH SHALL accept an Idempotency-Key header, and it SHALL be mandatory on all financially significant endpoints.	NOT IMPL.	Missing: No Idempotency-Key handling anywhere
FR-API-021	M	The System SHALL store the key, the request fingerprint, and the response for at least 30 days.	NOT IMPL.	Missing: No Idempotency-Key handling anywhere

FR-API-022	M	A repeated key with an identical request fingerprint SHALL return the stored response with Idempotent-Replay: true.	NOT IMPL.	Missing: No Idempotency-Key handling anywhere
FR-API-023	M	A repeated key with a different request fingerprint SHALL return 409 Conflict, because this indicates a client defect rather than a retry.	NOT IMPL.	Missing: No Idempotency-Key handling anywhere
NFR-API-001	M	The API SHALL be documented as an OpenAPI 3.1 specification generated from the implementation, so that documentation cannot drift from behaviour.	PARTIAL	Evidence: SwaggerModule wired at /docs, generated from decorators · Missing: Coverage thin; OpenAPI 3.0 not 3.1; not published or diffed
NFR-API-002	M	A deprecated endpoint or field SHALL be announced at least 180 days before removal, marked in the specification, and SHALL emit a Deprecation header.	NOT IMPL.	Missing: Not implemented.

Catalogue / Menu / Recipe — 42 requirements · 26% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
BR-MNU-001		A recipe SHALL NOT reference itself, directly or transitively, through sub-recipes. Cycle detection SHALL run on save and reject the change with the offending path displayed.	DESIGNED	Evidence: Designed in the Production Spec gate 7 and 11 · Missing: No implementation
BR-MNU-002		Publishing a new recipe version SHALL NOT alter the cost or consumption of previously completed orders. Historical orders reference recipe_version_id.	DESIGNED	Evidence: Designed in the Production Spec gate 7 and 11 · Missing: No implementation
BR-MNU-003		Recipe cost SHALL be computed as: $\text{cost}(\text{recipe}) = \sum \text{over lines of: } \text{quantity_in_base_unit} \times (1 + \text{wastage_percentage} / 100) \times \text{cost_per_base_unit}(\text{component}) \div (\text{yield_percentage} /$	DESIGNED	Evidence: Formula recorded; deferred · Missing: Costing deferred by D-17-05
BR-MNU-012		An item MAY be sold with an incomplete or absent recipe. The System SHALL permit the sale, SHALL record zero or partial cost, and SHALL list the item in a “recipes requiring c	NOT IMPL.	Missing: Requires Sales; not implemented
FR-MNU-001	M	The System SHALL support a menu hierarchy of: Menu → Category → Sub-category (optional) → Item → Variant.	COMPLETE	Evidence: Menu -> Category -> MenuItem -> MenuItemVariant models + routes; catalogue.e2e-spec.ts
FR-MNU-002	M	A tenant SHALL be able to define multiple menus (e.g. Breakfast, Main, Late Night, Ramadan, Delivery) and assign each to branches, order types, and time windows.	COMPLETE	Evidence: Multiple menus + menu_branches assignment; tested
FR-MNU-003	M	Where multiple menus are active simultaneously for a context, resolution SHALL follow explicit priority ordering, and the System SHALL warn on configuration that produces ambi	COMPLETE	Evidence: menus.service.resolveForBranch(): priority ordering + ambiguity warning; menu-resolution.spec.ts
FR-MNU-004	M	Menu items SHALL support the following attributes:	PARTIAL	Evidence: Item attributes modelled · Missing: tax_class_id nullable and FK-less (C-04); Fiscal out of scope
FR-MNU-005	M	Item names SHALL be independently configurable for each surface (POS, KDS, customer receipt, aggregator listing), because the appropriate name differs by context.	PARTIAL	Evidence: POS/kitchen/aggregator name columns · Missing: Receipt-surface name unmodelled in approved SQL
FR-MNU-006	M	Variants SHALL support independent pricing, independent recipes, independent barcodes, and independent availability.	PARTIAL	Evidence: Independent pricing, barcode, availability per variant · Missing: Independent recipes require Production Spec (not implemented)
FR-MNU-007	S	The System SHALL support item-level and category-level sort order, with drag-and-drop configuration and a live POS preview.	PARTIAL	Evidence: sort_order stored on items and categories · Missing: Drag-and-drop and live POS preview are UI, absent
FR-MNU-010	M	Modifier groups SHALL be defined once and attached to multiple items, with per-item overrides for price and default selection.	COMPLETE	Evidence: ModifierGroup/Modifier/ModifierGroupLink with min/max/required/repeat/default/free; tested
FR-MNU-011	M	Modifier groups SHALL support: minimum selections, maximum selections, required flag, allow-repeat flag, default selection, and free-quantity threshold (first two free, third	COMPLETE	Evidence: ModifierGroup/Modifier/ModifierGroupLink with min/max/required/repeat/default/free; tested
FR-MNU-012	M	Individual modifiers SHALL support linkage to a stock item and a consumption quantity, so their inventory impact is computed.	PARTIAL	Evidence: stock_item_id + consumption_unit_id recorded on modifiers · Missing: No FK and no inventory-impact computation
FR-MNU-013	S	Modifiers SHALL support a “recipe delta” specification, expressing changes to the parent recipe rather than an independent consumption: modifier: "Extra Cheese" price_delta: 5	PARTIAL	Evidence: recipe_delta stored as opaque JSONB · Missing: Never interpreted (C-11 / D-17-07); FR deferred · Depends on: D-17-07

FR-MNU-020	M	The System SHALL support named price lists, each with a scope (tenant, brand, branch, or branch group), a validity window, an optional recurrence schedule, and a priority.	PARTIAL	Evidence: PriceList scope/valid_from/valid_to/recurrence_rule/priority stored · Missing: No evaluation; branch_group scope deferred with ADR 0008 D-10
FR-MNU-021	M	The System SHALL support order-type-specific pricing, so delivery prices may differ from dine-in prices for the same item.	PARTIAL	Evidence: price_entries modelled · Missing: Order-type-specific resolution not implemented
FR-MNU-022	M	The System SHALL support time-based pricing with recurring schedules (e.g. weekdays 15:00–18:00), evaluated in the branch's timezone.	NOT IMPL.	Evidence: recurrence_rule column stored only · Missing: No time-window evaluation, no scheduling, no offline propagation
FR-MNU-023	M	Price changes SHALL be schedulable with a future effective date and SHALL propagate to offline terminals in advance of that date, activating locally at the correct local time.	NOT IMPL.	Evidence: recurrence_rule column stored only · Missing: No time-window evaluation, no scheduling, no offline propagation
FR-MNU-024	M	The System SHALL maintain full price change history: who changed what, from what to what, when, and effective when.	PARTIAL	Evidence: governance.audit_entries is the system of record (C-10) · Missing: No dedicated price-change history query surface
FR-MNU-025	S	The System SHALL support bulk price operations: percentage increase across a category, rounding to a configured price point (e.g. always end in .95), and CSV import with a pre	NOT IMPL.	Missing: No bulk price operations, no margin warnings
FR-MNU-026	S	The System SHALL warn when a price change would move an item's contribution margin below a configurable threshold, or below its cost.	NOT IMPL.	Missing: No bulk price operations, no margin warnings
FR-MNU-030	M	The System SHALL support manual availability toggling ("86-ing") per item, per branch, with an optional automatic re-enable time.	COMPLETE	Evidence: availability_rules + POST /catalogue/availability-rules/:id/86 with auto re-enable; tested
FR-MNU-031	M	The System SHALL automatically mark an item unavailable when any non-optional ingredient's stock reaches zero, if automatic availability is enabled for that branch.	NOT IMPL.	Evidence: org.branches.automatic_availability switch exists · Missing: Auto-86 needs Catalogue<->Inventory link, absent · Depends on: Inventory link
FR-MNU-032	M	Automatic unavailability SHALL be overridable by an authorised user, with the override recorded.	NOT IMPL.	Evidence: org.branches.automatic_availability switch exists · Missing: Auto-86 needs Catalogue<->Inventory link, absent · Depends on: Inventory link
FR-MNU-033	S	The System SHALL compute and display "remaining sellable quantity" per item, being the minimum across ingredients of (available stock ÷ required quantity per portion).	NOT IMPL.	Evidence: org.branches.automatic_availability switch exists · Missing: Auto-86 needs Catalogue<->Inventory link, absent · Depends on: Inventory link
FR-MNU-034	S	Availability state SHALL propagate to connected aggregator platforms within 60 seconds where the aggregator API supports it.	NOT IMPL.	Missing: No aggregator integration
FR-MNU-035	C	The System SHALL support daily quantity limits per item ("only 20 specials today"), decrementing on sale and auto-disabling at zero.	NOT IMPL.	Missing: Deferred by C-07; SRS [C] priority
FR-MNU-040	M	The System SHALL support recipes for menu item variants, for sub- recipes (prep items), and for central-kitchen production items.	DESIGNED	Evidence: docs/production/PRODUCTION_SPEC_DESIGN_GATE.md CLOSED/RATIFIED · Missing: Zero executable code: no production schema, model, migration, service, route or test · Depends on: Production Spec implementation
FR-MNU-041	M	Recipes SHALL specify components as either stock items or other recipes (sub-recipes), with quantity and unit.	DESIGNED	Evidence: docs/production/PRODUCTION_SPEC_DESIGN_GATE.md CLOSED/RATIFIED · Missing: Zero executable code: no production schema, model, migration, service, route or test · Depends on: Production Spec implementation
FR-MNU-042	M	The System SHALL detect and reject circular sub-recipe references, displaying the full cycle path.	DESIGNED	Evidence: docs/production/PRODUCTION_SPEC_DESIGN_GATE.md CLOSED/RATIFIED · Missing: Zero executable code: no production schema, model, migration, service, route or test · Depends on: Production Spec implementation
FR-MNU-043	M	Recipes SHALL support a yield specification: the quantity produced, and a yield percentage representing preparation loss.	DESIGNED	Evidence: docs/production/PRODUCTION_SPEC_DESIGN_GATE.md CLOSED/RATIFIED · Missing: Zero executable code: no production schema, model, migration, service, route or test · Depends on: Production Spec implementation
FR-MNU-044	M	Recipe lines SHALL support a per-component wastage percentage representing trim loss (e.g. peeling potatoes loses 18%).	DESIGNED	Evidence: docs/production/PRODUCTION_SPEC_DESIGN_GATE.md CLOSED/RATIFIED · Missing: Zero executable code: no production schema, model, migration, service, route or test · Depends on: Production Spec implementation

FR-MNU-045	M	Recipes SHALL be versioned. Publishing a new version SHALL supersede but not delete the prior version, and completed orders SHALL retain their reference to the version in forc	DESIGNED	Evidence: docs/production/PRODUCTION_SPEC_DESIGN_GATE.md CLOSED/RATIFIED · Missing: Zero executable code: no production schema, model, migration, service, route or test · Depends on: Production Spec implementation
FR-MNU-046	M	Recipe cost SHALL be automatically recomputed when any component's valuation changes, and the recomputation SHALL cascade through dependent sub-recipes and parent recipes.	DESIGNED	Evidence: Design gate records it as deferred · Missing: Costing deferred by D-17-05; no implementation · Depends on: D-17-05 + Inventory valuation
FR-MNU-047	S	The System SHALL support recipe scoping, allowing a branch to hold a variant recipe differing from the brand standard, with the deviation visible in a compliance report.	DESIGNED	Evidence: Scope model designed (D-17-03) · Missing: No implementation; compliance report excluded from scope
FR-MNU-048	S	The System SHALL support recipe scaling: displaying a recipe scaled to an arbitrary output quantity for batch production.	OUT OF SCOPE	Evidence: Design gate 20 excludes scaling and nutrition · Depends on: Explicit scope boundary
FR-MNU-049	S	The System SHALL store localised preparation instructions and reference images per recipe, viewable from the KDS.	DESIGNED	Evidence: docs/production/PRODUCTION_SPEC_DESIGN_GATE.md CLOSED/RATIFIED · Missing: Zero executable code: no production schema, model, migration, service, route or test · Depends on: Production Spec implementation
FR-MNU-050	C	The System SHALL compute nutritional information per portion by aggregating component nutritional data where supplied. This is informational only and is explicitly not certifi	OUT OF SCOPE	Evidence: Design gate 20 excludes scaling and nutrition · Depends on: Explicit scope boundary
FR-MNU-055	S	The System SHALL classify menu items on the Boston-matrix menu engineering model using popularity (units sold relative to menu average) and profitability (contribution margin)	NOT IMPL.	Missing: Menu engineering needs Sales data; not implemented
FR-MNU-056	S	The classification SHALL be computable per branch, per day-part, and per date range.	NOT IMPL.	Missing: Menu engineering needs Sales data; not implemented
FR-MNU-057	C	The System SHALL analyse modifier attachment rates and suggest price adjustments for modifiers whose attachment rate exceeds a threshold (indicating under- pricing) or falls b	NOT IMPL.	Missing: Menu engineering needs Sales data; not implemented

Core Money & Units — 4 requirements · 25% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
BR-CORE-001		Arithmetic between different currencies SHALL raise an error. There is no implicit conversion.	NOT IMPL.	Missing: No Money value type, no currency-mismatch guard, no allocate() implementation
BR-CORE-002		allocate SHALL distribute an amount across ratios such that the sum of the results exactly equals the input, distributing remainder minor units to the largest ratios first.	NOT IMPL.	Missing: No Money value type, no currency-mismatch guard, no allocate() implementation
BR-CORE-003		Quantities SHALL be stored with 6 decimal places of precision.	COMPLETE	Evidence: NUMERIC(18,6) on every quantity column; verified in migrations
BR-CORE-004		Conversion between units of different dimensions (mass ↔ volume) SHALL require an item-specific density factor and SHALL fail if none is configured for that item.	NOT IMPL.	Evidence: inventory.uom_conversions table exists · Missing: No density factor logic; cross-dimension conversion not implemented

NFR - Data — 2 requirements · 25% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
NFR-DATA-001	M	Stock movement records SHALL be retained for the statutory period of the branch's jurisdiction or 7 years, whichever is greater.	PARTIAL	Evidence: Monthly partitions retained, never dropped; no DELETE grant · Missing: No retention enforcement or archival to object storage
NFR-DATA-002	M	Aggregated reporting data SHALL be no more than 15 minutes behind the transactional source during trading hours.	NOT IMPL.	Missing: Not implemented.

Security & Authorization — 45 requirements · 20% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-SEC-001	M	The System SHALL implement role-based access control where permissions are granted to roles and roles are assigned to users.	COMPLETE	Evidence: roles/permissions/role_permissions/membership_roles + /auth/roles"; rbac.e2e-spec.ts
FR-SEC-002	M	Role assignments SHALL carry a scope, restricting the assignment to a tenant, a brand, a set of branches, or a single branch.	NOT IMPL.	Evidence: membership_roles.branch_id column exists but is unused · Missing: Scoped assignment not implemented; ADR 0008 D-02 deferred · Depends on: ADR 0008 D-02

FR-SEC-003	M	A user MAY hold multiple role assignments with different scopes — for example, Branch Manager at Branch 1 and Cashier at Branch 2.	PARTIAL	Evidence: Multiple role assignments per membership supported · Missing: Assignments carry no scope
FR-SEC-004	M	Where multiple assignments apply, effective permissions SHALL be the union of granted permissions within each assignment's own scope. Permissions SHALL NOT leak across scopes.	PARTIAL	Evidence: Permission union computed per membership; rbac tests · Missing: No per-scope isolation; leak prevention untestable without scopes
FR-SEC-005	S	Role assignments SHALL support validity dates, enabling temporary elevation (covering a manager's leave) that expires automatically.	NOT IMPL.	Missing: No validity dates on membership_roles
FR-SEC-010	M	The System SHALL ship predefined roles suitable for immediate use, which tenants may clone and modify but not delete.	NOT IMPL.	Missing: No predefined/system role set shipped or seeded
FR-SEC-011	M	Tenants SHALL be able to create custom roles by selecting permissions from the catalogue.	COMPLETE	Evidence: POST /auth/roles + POST /auth/roles/:id/permissions from permission catalogue; tested
FR-SEC-012	M	The role editor SHALL display, for each permission, a plain-language description of what it allows and a warning marker on sensitive permissions. 15.4 Segregation of Duties	PARTIAL	Evidence: Permission catalogue carries description + module · Missing: No sensitive-permission warning marker
FR-SEC-015	M	The System SHALL define incompatible permission pairs and SHALL warn when a role or user combines them.	NOT IMPL.	Missing: No SoD mechanism; SRS 15.4 defines pairs but no enforcement exists
FR-SEC-016	M	The System SHALL block, not merely warn, on the following combinations regardless of role configuration: approving one's own requisition, approving one's own discount, approvi	NOT IMPL.	Missing: No SoD mechanism; SRS 15.4 defines pairs but no enforcement exists
FR-SEC-017	S	The System SHALL provide an SoD conflict report listing every user whose effective permissions contain an incompatible pair.	NOT IMPL.	Missing: No SoD mechanism; SRS 15.4 defines pairs but no enforcement exists
FR-SEC-020	M	The System SHALL support the following authentication methods:	PARTIAL	Evidence: Email+password with JWT; terminal-bound sessions · Missing: PIN and MFA methods absent
FR-SEC-021	M	PIN authentication SHALL be valid only on registered terminals within the employee's permitted branches, and SHALL NOT grant access to the web dashboard.	NOT IMPL.	Missing: No PIN authentication, no PIN hashing/lockout
FR-SEC-022	M	PINs SHALL be stored as salted hashes, SHALL be unique within a branch, and SHALL be subject to lockout after a configurable number of failures.	NOT IMPL.	Missing: No PIN authentication, no PIN hashing/lockout
FR-SEC-023	M	Multi-factor authentication SHALL be supported for dashboard access and SHALL be enforceable as mandatory per role.	NOT IMPL.	Missing: No MFA/TOTP
FR-SEC-024	M	MFA SHALL be mandatory for any role holding security.user.manage, settings.tenant.manage, or api.key.manage.	NOT IMPL.	Missing: No MFA/TOTP
FR-SEC-025	M	Password policy SHALL be configurable per tenant with a minimum of 10 characters, and SHALL check candidate passwords against a known-breached-password list.	PARTIAL	Evidence: assertPasswordMeetsPolicy() enforces minimum length · Missing: Not per-tenant configurable; no breached-password list check
FR-SEC-026	M	Sessions SHALL expire after a configurable idle period: default 15 minutes on POS, 60 minutes on dashboard, 8 hours on KDS.	PARTIAL	Evidence: JWT_ACCESS_TTL=15m / JWT_REFRESH_TTL=30d configurable · Missing: No per-surface idle expiry (POS 15m / dashboard 60m / KDS 8h)
FR-SEC-027	M	The System SHALL support forced logout of a user's sessions across all devices by an administrator.	PARTIAL	Evidence: Session revocation on password change/reset (ADR 0005), tested · Missing: No administrator-initiated forced-logout surface
FR-SEC-028	M	Terminals SHALL be individually registered, and the System SHALL support revoking a terminal's registration, immediately invalidating its credentials and wiping its local data	PARTIAL	Evidence: Terminal registration + fingerprints + status endpoints (ADR 0004) · Missing: No credential wipe on revocation; pairing/activation deferred
FR-SEC-030	M	The System SHALL provide a general approval mechanism used by discounts, refunds, purchase orders, waste, count adjustments, expenses, and price changes.	NOT IMPL.	Evidence: Inventory exposes a requires_approval gate only (B-2) · Missing: No general approval mechanism, request entity, escalation, or offline policy
FR-SEC-031	M	Approval requests SHALL specify: the requesting user, the action, the affected entity, the value, the required approver permission, and an expiry.	NOT IMPL.	Evidence: Inventory exposes a requires_approval gate only (B-2) · Missing: No general approval mechanism, request entity, escalation, or offline policy
FR-SEC-032	M	Approvals SHALL be obtainable synchronously (manager PIN on the terminal) or asynchronously (push notification to the manager's mobile device), with the terminal remaining usa	NOT IMPL.	Evidence: Inventory exposes a requires_approval gate only (B-2) · Missing: No general approval mechanism, request entity, escalation, or offline policy

FR-SEC-033	M	Approval decisions SHALL record approver, timestamp, decision, and any comment, and SHALL be immutable.	NOT IMPL.	Evidence: Inventory exposes a requires_approval gate only (B-2) · Missing: No general approval mechanism, request entity, escalation, or offline policy
FR-SEC-034	S	The System SHALL support escalation: if no decision is made within a configured period, the request escalates to the next approval level.	NOT IMPL.	Evidence: Inventory exposes a requires_approval gate only (B-2) · Missing: No general approval mechanism, request entity, escalation, or offline policy
FR-SEC-035	M	Where an operation must proceed offline and no approver is present, the System SHALL support a configurable policy: block the operation, or permit it with mandatory retrospect	NOT IMPL.	Evidence: Inventory exposes a requires_approval gate only (B-2) · Missing: No general approval mechanism, request entity, escalation, or offline policy
FR-SEC-040	M	All network traffic SHALL use TLS 1.3 with TLS 1.2 as the minimum fallback.	NOT IMPL.	Missing: No TLS config, at-rest encryption, envelope encryption, or KMS in repo
FR-SEC-041	M	All data at rest SHALL be encrypted: database volumes, object storage, backups, and the POS local database.	NOT IMPL.	Missing: No TLS config, at-rest encryption, envelope encryption, or KMS in repo
FR-SEC-042	M	Sensitive fields (national identifiers, bank details, API secrets) SHALL be encrypted at the application layer with envelope encryption and per-tenant data keys, in addition t	NOT IMPL.	Missing: No TLS config, at-rest encryption, envelope encryption, or KMS in repo
FR-SEC-043	M	Encryption keys SHALL be managed by a managed KMS, SHALL be rotated at least annually, and SHALL never be committed to source control or logged.	NOT IMPL.	Missing: No TLS config, at-rest encryption, envelope encryption, or KMS in repo
FR-SEC-044	M	The System SHALL NOT store card PANs, CVV, magnetic stripe data, or PIN blocks anywhere, including logs, backups, and crash reports.	NOT IMPL.	Missing: Vacuously held - no payment domain exists. No enforcement control implemented.
FR-SEC-045	M	Every API endpoint SHALL enforce authorisation server-side. Client-side permission checks are presentation only and SHALL NOT be relied upon.	COMPLETE	Evidence: JwtAuthGuard + PermissionsGuard on every business route; 114 routes audited; 403/404 tested
FR-SEC-046	M	The System SHALL apply rate limiting and progressive lockout on authentication endpoints.	PARTIAL	Evidence: AuthThrottlerGuard on auth endpoints; throttle.e2e-spec.ts · Missing: No progressive lockout
FR-SEC-047	M	All user input SHALL be validated against an explicit schema at the API boundary, rejecting unknown fields rather than ignoring them.	COMPLETE	Evidence: Global ValidationPipe whitelist + forbidNonWhitelisted + transform; rejection tested
FR-SEC-048	M	Database access SHALL use parameterised queries exclusively; string- concatenated SQL is prohibited and SHALL fail static analysis.	PARTIAL	Evidence: Prisma parameterised queries; 1 \$queryRawUnsafe in reconciliation uses bound params · Missing: No static-analysis gate enforcing the prohibition
FR-SEC-049	M	Dependencies SHALL be scanned for known vulnerabilities on every build, and builds SHALL fail on critical findings.	NOT IMPL.	Missing: No CI dependency scanning, no secret manager, no pen-test process
FR-SEC-050	M	Secrets SHALL be injected at runtime from a secret manager, never present in images or repositories.	NOT IMPL.	Missing: No CI dependency scanning, no secret manager, no pen-test process
FR-SEC-051	S	The System SHALL undergo an external penetration test before general availability and annually thereafter.	NOT IMPL.	Missing: No CI dependency scanning, no secret manager, no pen-test process
FR-SEC-052	M	The System SHALL support IP allow-listing for API access and for dashboard access at the Enterprise tier.	NOT IMPL.	Missing: No IP allow-listing, no SIEM sink
FR-SEC-053	M	Security-relevant events SHALL be forwarded to a SIEM-compatible sink, and Enterprise tenants SHALL be able to receive their own security event stream.	NOT IMPL.	Missing: No IP allow-listing, no SIEM sink
FR-SEC-060	M	The System SHALL classify data by sensitivity and apply corresponding controls:	NOT IMPL.	Missing: No data classification, retention job, DSR handling, residency, or backup verification
FR-SEC-061	M	Data retention periods SHALL be configurable within statutory bounds, and a scheduled job SHALL purge or anonymise data past retention.	NOT IMPL.	Missing: No data classification, retention job, DSR handling, residency, or backup verification
FR-SEC-062	M	The System SHALL support data subject requests: access (export), rectification, and erasure (anonymisation preserving financial records).	NOT IMPL.	Missing: No data classification, retention job, DSR handling, residency, or backup verification
FR-SEC-063	M	Tenants SHALL be able to select a data residency region at signup, and their data SHALL remain within that region including backups.	NOT IMPL.	Missing: No data classification, retention job, DSR handling, residency, or backup verification
FR-SEC-064	M	Backups SHALL be encrypted, tested by restoration at least quarterly, and retained per a documented schedule.	NOT IMPL.	Missing: No data classification, retention job, DSR handling, residency, or backup verification

Data Retention & Migrations — 10 requirements · 15% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-DR-001	M	High-volume, time-ordered tables SHALL be range-partitioned by time:	PARTIAL	Evidence: inventory.stock_movements RANGE-partitioned by occurred_at, 14 monthly partitions · Missing: Other high-volume tables (orders, audit) not partitioned
FR-DR-002	M	Partitions SHALL be created automatically at least 3 months in advance by a scheduled job, with alerting if creation fails.	NOT IMPL.	Missing: Explicitly deferred; no scheduler exists. Partition creation is a manual step.
FR-DR-003	M	Archived partitions SHALL be exported to object storage in a queryable columnar format and SHALL remain retrievable within the statutory retention window.	NOT IMPL.	Missing: No archival to object storage
FR-DR-010	M	All schema changes SHALL be expressed as versioned, forward-only migrations checked into source control.	COMPLETE	Evidence: 13 versioned forward-only Prisma migrations in source control; migrate status clean
FR-DR-011	M	Migrations SHALL be backward compatible with the previously deployed application version, enabling zero-downtime deployment and rollback.	NOT IMPL.	Missing: No documented/rehearsed process, staging dataset, or per-tenant restore
FR-DR-012	M	Breaking schema changes SHALL follow the expand-migrate-contract sequence: 1. Expand : add the new column/table, nullable, with no constraint. Deploy. 2. Backfill : populate	NOT IMPL.	Missing: No documented/rehearsed process, staging dataset, or per-tenant restore
FR-DR-013	M	Migrations SHALL be tested against a production-sized dataset in staging with timing measured, and any migration exceeding a 5-second lock SHALL be rejected.	NOT IMPL.	Missing: No documented/rehearsed process, staging dataset, or per-tenant restore
FR-DR-014	M	Adding a NOT NULL column to a large table SHALL use a nullable column plus a backfill plus a validated constraint, never a single blocking statement.	NOT IMPL.	Missing: No documented/rehearsed process, staging dataset, or per-tenant restore
FR-DR-020	M	Backups SHALL be encrypted, and restore procedures SHALL be documented and rehearsed quarterly with recorded results.	NOT IMPL.	Missing: No documented/rehearsed process, staging dataset, or per-tenant restore
FR-DR-021	M	The System SHALL support restoring a single tenant's data without restoring the entire database, using logical export filtered by tenant_id.	NOT IMPL.	Missing: No documented/rehearsed process, staging dataset, or per-tenant restore

Branch Ops & Central Kitchen — 30 requirements · 12% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-BRN-001	M	The System SHALL support an unlimited number of branches per brand and an unlimited number of brands per tenant, subject to plan limits.	COMPLETE	Evidence: No cardinality limits in schema; brands/branches CRUD + tests
FR-BRN-002	M	Each branch SHALL hold its own: inventory, cash drawers, staff roster, operating hours, timezone, currency, tax configuration, and country pack assignment.	PARTIAL	Evidence: Branch holds operating hours, timezone, base currency, country code; stock held per org.locations · Missing: Cash drawers and staff roster domains absent
FR-BRN-003	M	Branches in different countries SHALL operate under different country packs within one tenant, and consolidated reporting SHALL handle multi-currency correctly.	PARTIAL	Evidence: branches.country_code recorded · Missing: No country packs; no consolidation
FR-BRN-004	M	Consolidated multi-currency reports SHALL display both the original currency amounts and a converted total, with the conversion rate, rate source, and rate date shown explicit	NOT IMPL.	Missing: Not implemented.
FR-BRN-005	M	The System SHALL support branch groups (regions, clusters, franchise territories) as a reporting and permission-scoping dimension.	NOT IMPL.	Missing: Branch groups deferred by ADR 0008 D-10 · Depends on: ADR 0008 D-10
FR-BRN-006	S	The System SHALL support central menu and price management with per-branch override capability, and the System SHALL report which branches deviate from the brand standard.	NOT IMPL.	Missing: Not implemented.
FR-BRN-007	S	The System SHALL support central recipe management with the same override-and-report pattern.	NOT IMPL.	Missing: Not implemented.
FR-BRN-008	S	The System SHALL support a branch template: a new branch may be created by copying configuration (menu, stations, roles, settings, printers) from an existing branch.	NOT IMPL.	Missing: Not implemented.

FR-BRN-010	S	The System SHALL provide a branch scorecard comparing branches on: net sales, transaction count, average order value, food cost %, labour cost %, prime cost %, waste %, varian	NOT IMPL.	Missing: Not implemented.
FR-BRN-011	S	Comparison SHALL support normalisation by seat count, floor area, labour hours, and trading hours, so branches of different sizes are meaningfully comparable.	NOT IMPL.	Missing: Not implemented.
FR-BRN-012	S	The System SHALL rank branches on any metric and SHALL indicate movement against the prior period.	NOT IMPL.	Missing: Not implemented.
FR-BRN-013	S	The System SHALL highlight outliers, defined as branches deviating beyond a configurable number of standard deviations from the group mean on any tracked metric.	NOT IMPL.	Missing: Not implemented.
FR-BRN-014	C	The System SHALL support like-for-like comparison, excluding branches open for less than a configurable maturity period from group averages.	NOT IMPL.	Missing: Not implemented.
FR-BRN-015	M	The System SHALL support stock transfers between any two locations within a tenant, per FR-INV-031.	COMPLETE	Evidence: POST /inventory/transfers + /receive with counterpart pairing; e2e tested
FR-BRN-016	S	The System SHALL support transfer requests: a branch requests stock, the source location approves and dispatches.	NOT IMPL.	Missing: Not implemented.
FR-BRN-017	S	The System SHALL suggest transfers where one location holds excess stock approaching expiry and another location shows a shortage of the same item.	NOT IMPL.	Missing: Not implemented.
FR-BRN-020	S	The System SHALL model central kitchens as locations that consume input stock and produce output stock.	PARTIAL	Evidence: org.central_kitchens modelled and registered in org.locations · Missing: Consume/produce production flow not implemented · Depends on: Production Spec
FR-BRN-021	S	Production SHALL be driven by a production order specifying output item, target quantity, and target date.	BLOCKED	Missing: Production orders require Production Spec recipes · Depends on: Production Spec
FR-BRN-022	S	A production order SHALL expand its recipe to compute required inputs, SHALL check availability, and SHALL flag shortages before production begins.	BLOCKED	Missing: Production orders require Production Spec recipes · Depends on: Production Spec
FR-BRN-023	S	On completion, a production order SHALL record actual output quantity, consume input stock, produce output stock, and create an output batch with production date and expiry.	BLOCKED	Missing: Production orders require Production Spec recipes · Depends on: Production Spec
FR-BRN-024	S	The System SHALL compute production yield variance: $\text{Yield Variance} = \text{Actual Output} - (\text{Theoretical Output from Actual Inputs})$	BLOCKED	Missing: Production orders require Production Spec recipes · Depends on: Production Spec
FR-BRN-025	S	Production output cost SHALL be computed as the sum of consumed input costs plus configurable production overhead (labour, energy) per unit or per batch.	BLOCKED	Missing: Production orders require Production Spec recipes · Depends on: Production Spec
FR-BRN-026	S	The System SHALL generate a production plan from branch demand forecasts and current branch stock, indicating what the central kitchen should produce for the next production c	BLOCKED	Missing: Production orders require Production Spec recipes · Depends on: Production Spec
FR-BRN-027	S	The System SHALL support distribution orders allocating produced quantities to branches, generating transfer movements and, where relevant, delivery manifests.	BLOCKED	Missing: Production orders require Production Spec recipes · Depends on: Production Spec
FR-BRN-028	S	Where produced quantity is insufficient for all branch requests, the System SHALL support allocation rules: proportional, by priority ranking, or manual.	BLOCKED	Missing: Production orders require Production Spec recipes · Depends on: Production Spec
FR-BRN-029	S	The System SHALL maintain full batch traceability from raw material receipt through production to the branch and to the customer order.	BLOCKED	Missing: Production orders require Production Spec recipes · Depends on: Production Spec
FR-BRN-030	C	The System SHALL support internal transfer pricing between central kitchen and branch, enabling the central kitchen to be evaluated as a cost centre or a profit centre per ten	BLOCKED	Missing: Production orders require Production Spec recipes · Depends on: Production Spec
FR-BRN-035	C	The System SHALL support franchise branches with restricted configuration authority: franchisees may operate but may not modify brand-controlled menu, recipes, or pricing.	NOT IMPL.	Missing: Not implemented.

FR-BRN-036	C	The System SHALL compute franchise royalties as a configurable percentage of net sales, with reporting per franchisee per period.	NOT IMPL.	Missing: Not implemented.
FR-BRN-037	C	The System SHALL provide franchisor visibility into franchisee operational compliance: menu deviation, price deviation, recipe deviation, and mandated-supplier usage.	NOT IMPL.	Missing: Not implemented.

NFR - Observability — 7 requirements · 7% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
NFR-OBS-001		All logs structured (JSON), with tenant, branch, correlation, and causation identifiers	PARTIAL	Evidence: audit_entries carries correlation_id · Missing: No structured JSON log pipeline, no request correlation middleware
NFR-OBS-002		Distributed tracing across API, workers, and database	NOT IMPL.	Missing: No logging pipeline, tracing, metrics or alerting.
NFR-OBS-003		RED metrics (Rate, Errors, Duration) per endpoint and per handler	NOT IMPL.	Missing: No logging pipeline, tracing, metrics or alerting.
NFR-OBS-004		Business metrics emitted: orders/min, sync backlog, fiscal failures, offline terminals	NOT IMPL.	Missing: No logging pipeline, tracing, metrics or alerting.
NFR-OBS-005		No PII or secrets in logs, enforced by a redaction layer with an allowlist	NOT IMPL.	Missing: No logging pipeline, tracing, metrics or alerting.
NFR-OBS-006		Alerts defined for every SLO breach with documented runbooks	NOT IMPL.	Missing: No logging pipeline, tracing, metrics or alerting.
NFR-OBS-007		Per-tenant health view available to support without database access	NOT IMPL.	Missing: No logging pipeline, tracing, metrics or alerting.

NFR - Performance — 16 requirements · 3% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
NFR-PERF-001	M	Item addition to an order SHALL render within 100 ms at the 95th percentile on the reference device (Android 11, 4 GB RAM, quad-core 1.8 GHz).	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-002	M	Payment finalisation, including local persistence and print job dispatch, SHALL complete within 800 ms at p95 offline, and 1,500 ms at p95 online.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-003	M	POS cold start SHALL complete within 6 seconds on the reference device.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-004	M	A fired order SHALL appear on the target station display within 1 second at p95 on the local network.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-005	M	Stock level queries for a single location with up to 3,000 items SHALL return within 500 ms at p95.	PARTIAL	Evidence: stock_levels carries tenant_id + index (tenant_id, location_id, stock_item_id) · Missing: Never measured; no performance test exists
NFR-PERF-006	M	Recipe expansion and inventory depletion for a completed order of up to 30 lines SHALL complete within 200 ms at p95 and SHALL execute within the order's transaction.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-010	M	Standard reports over a 31-day period for a single branch SHALL render within 2 seconds at p95.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-011	M	Consolidated reports across up to 100 branches over a 31-day period SHALL render within 5 seconds at p95.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-012	M	Dashboard initial load SHALL complete within 3 seconds at p95.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-020	M	A backlog of 5,000 queued operations SHALL synchronise within 5 minutes on a 2 Mbps connection.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-021	M	Local order persistence SHALL complete within 50 ms at p95 on the reference device.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-030	M	Read endpoints SHALL respond within 200 ms at p95 and 500 ms at p99, excluding reporting endpoints.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-031	M	Write endpoints SHALL respond within 400 ms at p95.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.

NFR-PERF-032	M	The sync batch endpoint SHALL process 500 operations within 3 seconds at p95.	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-040		Web dashboard Largest ≤ 2.5 s Contentful Paint	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.
NFR-PERF-041		Web dashboard Interaction ≤ 200 ms to Next Paint	NOT IMPL.	Missing: No performance test, benchmark or measurement harness exists.

Localisation & Country Packs — 20 requirements · 2% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-L0C-001	M	The System SHALL support Arabic and English as first-class languages across every surface: POS, KDS, dashboard, mobile, receipts, tickets, reports, exports, emails, and error	NOT IMPL.	Missing: All are UI/rendering/printing requirements; no client surface exists in this repository
FR-L0C-002	M	The Arabic interface SHALL use clear Modern Standard Arabic, not a regional dialect.	NOT IMPL.	Missing: All are UI/rendering/printing requirements; no client surface exists in this repository
FR-L0C-003	M	Layout direction SHALL follow the selected language: full RTL mirroring for Arabic, LTR for English, applied to layout, iconography with directional meaning, progress indicato	NOT IMPL.	Missing: All are UI/rendering/printing requirements; no client surface exists in this repository
FR-L0C-004	M	The System SHALL correctly render bidirectional text: an Arabic dish name containing a Latin brand name, or an English item name containing an Arabic word, SHALL render with c	NOT IMPL.	Missing: All are UI/rendering/printing requirements; no client surface exists in this repository
FR-L0C-005	M	Numeral rendering SHALL be configurable between Western (0123456789) and Arabic-Indic (٠١٢٣٤٥٦٧٨٩), independently for the interface and for printed documents.	NOT IMPL.	Missing: All are UI/rendering/printing requirements; no client surface exists in this repository
FR-L0C-006	M	All user-authored content — item names, descriptions, categories, modifiers, reason codes, station names, expense categories — SHALL be storable in every enabled language, wit	PARTIAL	Evidence: Localised JSONB name/label columns across catalogue and inventory · Missing: No translation management; no locale negotiation
FR-L0C-007	M	Where a translation is missing, the System SHALL display the fallback language content rather than an empty string or a key.	NOT IMPL.	Missing: No fallback-language resolution implemented
FR-L0C-008	M	Language SHALL be selectable per user, per terminal, and per printed document type independently. A branch may run an English-language dashboard, an Arabic POS, an Urdu kitche	NOT IMPL.	Missing: All are UI/rendering/printing requirements; no client surface exists in this repository
FR-L0C-009	S	The System SHALL support additional languages by translation pack: French, Turkish, Urdu, Bengali, Hindi, Tagalog.	NOT IMPL.	Missing: All are UI/rendering/printing requirements; no client surface exists in this repository
FR-L0C-010	M	Dates, times, numbers, and currency SHALL be formatted per locale, and the System SHALL support both Gregorian and Hijri calendar display where the country pack enables it.	NOT IMPL.	Missing: All are UI/rendering/printing requirements; no client surface exists in this repository
FR-L0C-011	M	Arabic typography SHALL use fonts selected and tested for legibility at small sizes on thermal printers and at 2-metre viewing distance on kitchen displays.	NOT IMPL.	Missing: All are UI/rendering/printing requirements; no client surface exists in this repository
FR-L0C-012	M	Thermal printing of Arabic SHALL be tested across the supported printer matrix, with the System supporting both native Arabic font rendering and image-based rendering as a fal	NOT IMPL.	Missing: All are UI/rendering/printing requirements; no client surface exists in this repository
FR-L0C-020	M	All jurisdiction-specific behaviour SHALL be driven by the country pack. No country-specific logic SHALL be compiled into core application code.	NOT IMPL.	Evidence: tenants.country_pack_code is a recorded string · Missing: No country pack entity, versioning, signing, conformance suite or tax engine registry
FR-L0C-021	M	Country packs SHALL be versioned with effective dates, and historical transactions SHALL be interpreted under the pack version in force at their transaction time.	NOT IMPL.	Evidence: tenants.country_pack_code is a recorded string · Missing: No country pack entity, versioning, signing, conformance suite or tax engine registry
FR-L0C-022	M	Country packs SHALL be cryptographically signed, and the System SHALL reject an unsigned or invalidly-signed pack.	NOT IMPL.	Evidence: tenants.country_pack_code is a recorded string · Missing: No country pack entity, versioning, signing, conformance suite or tax engine registry
FR-L0C-023	M	Country packs SHALL be validated against a conformance test suite before activation, covering tax computation, rounding, invoice field completeness, and QR generation.	NOT IMPL.	Evidence: tenants.country_pack_code is a recorded string · Missing: No country pack entity, versioning, signing, conformance suite or tax engine registry

FR-L0C-024	M	Country packs SHALL be distributed to offline terminals in advance of their effective date and SHALL activate locally by date.	NOT IMPL.	Evidence: tenants.country_pack_code is a recorded string · Missing: No country pack entity, versioning, signing, conformance suite or tax engine registry
FR-L0C-025	M	The tax engine referenced by a pack SHALL be one of a registered set of strategy implementations. Adding a genuinely novel tax model requires a new strategy implementation; ch	NOT IMPL.	Evidence: tenants.country_pack_code is a recorded string · Missing: No country pack entity, versioning, signing, conformance suite or tax engine registry
FR-L0C-030	S	The System SHALL provide a country pack authoring and validation tool so that a new pack can be produced, tested against the conformance suite, and certified without core code	NOT IMPL.	Evidence: tenants.country_pack_code is a recorded string · Missing: No country pack entity, versioning, signing, conformance suite or tax engine registry
FR-L0C-031	M	A country pack SHALL NOT be activatable in production until it passes the full conformance suite and is signed by an authorised release key.	NOT IMPL.	Evidence: tenants.country_pack_code is a recorded string · Missing: No country pack entity, versioning, signing, conformance suite or tax engine registry

Sales / POS — 80 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
BR-P0S-001		An Order in state COMPLETED SHALL NOT be modified. Corrections are made by creating a Refund referencing it.	NOT IMPL.	Missing: Sales/POS domain does not exist
BR-P0S-002		An Order SHALL NOT transition to COMPLETED while paid_total + discount_total_of_comps < grand_total.	NOT IMPL.	Missing: Sales/POS domain does not exist
BR-P0S-003		An Order SHALL NOT transition to CANCELLED if any line has been fired to the kitchen and bumped, unless a user with the order.cancel_after_production permission approves and a	NOT IMPL.	Missing: Sales/POS domain does not exist
BR-P0S-004		item_name_snapshot, unit_price, tax_class_id, unit_cost_snapshot, and recipe_version_id SHALL be captured at the time of sale and SHALL NOT be recomputed from current master d	NOT IMPL.	Missing: Sales/POS domain does not exist
FR-P0S-001	M	The System SHALL allow creation of an order in the following types: dine- in, takeaway, delivery, drive-through, pickup, and aggregator-injected.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-002	M	The System SHALL assign a human-readable order number, unique within a branch within a business day, generated locally without requiring server connectivity. Order number form	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-003	M	For dine-in orders, the System SHALL require table assignment before firing to the kitchen.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-004	S	The System SHALL support assigning order lines to seat numbers within a dine-in order, to enable seat-level splitting and service.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-005	M	The System SHALL allow multiple orders to be open simultaneously on one terminal and SHALL allow switching between them without data loss.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-006	M	The System SHALL allow an order to be parked (set aside) and resumed by the same or another authorised user on the same or another terminal within the branch.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-007	M	The System SHALL record opened_by, served_by, and closed_by employee identities on every order.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-010	M	The System SHALL present the menu as a configurable grid of categories and items, with per-branch layout, colour coding, and item images.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-011	M	The System SHALL support item lookup by: category navigation, text search (Arabic and English, diacritic- and hamza-insensitive), barcode scan, PLU code, and a configurable fa	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-012	M	Arabic search SHALL normalise ﺍﺀﺍﺀ to a single form, ﺍﺀ to ﺍﺀ, ﺍﺀ to ﺍﺀ, and SHALL ignore tashkeel, so that a cashier typing "شاورما" finds "شاورما".	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-013	M	The System SHALL support items with multiple variants (size, preparation) and SHALL require variant selection before the line is added.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.

FR-P0S-014	M	The System SHALL support fractional quantities for items sold by weight, with input from a connected scale where available.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-015	M	The System SHALL support open-price items where the price is entered at sale time, restricted by permission and bounded by configurable min/max.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-016	M	The System SHALL support open-description items (“miscellaneous”) mapped to a configured revenue category and tax class.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-020	M	The System SHALL present modifier groups attached to an item immediately upon selection, in configured order, enforcing minimum and maximum selection counts.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-021	M	Modifiers SHALL support three semantic kinds:	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-022	M	Modifier price deltas SHALL be configurable per price list, per branch, and per order type. A modifier may be free for dine-in and charged for delivery.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-023	S	The System SHALL support nested modifier groups to a depth of 2 (e.g. “Add sauce” → “which sauce” → “how much”).	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-024	M	Removal modifiers SHALL reduce ingredient consumption in the inventory depletion calculation. A “no cheese” burger SHALL NOT deplete cheese.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-025	S	The System SHALL support free-text kitchen notes per line, with a configurable per-branch option to disable them.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-026	C	The System SHALL support a library of predefined note chips (“no ice”, “well done”, “separate packaging”) to reduce free-text usage.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-030	S	The System SHALL support combo items composed of slots, where each slot offers a choice among configured options, with optional price deltas for premium choices.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-031	S	Combo pricing SHALL support fixed price, sum-of-components-minus- discount, and component-price-override strategies.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-032	S	Combo revenue SHALL be allocable to component items for reporting purposes, using a configurable allocation basis (equal, by list price, by cost).	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-035	M	The System SHALL support firing an order to the kitchen either automatically upon line entry (fast-casual mode) or explicitly by user action (table-service mode), configurable	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-036	S	The System SHALL support courses, allowing lines to be grouped and fired independently in sequence.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-037	S	The System SHALL support a “hold and fire” instruction where a course is sent to the kitchen but marked not to start until released.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-038	M	Adding a line to an already-fired order SHALL create an amendment ticket to the kitchen clearly marked as an addition, not a reprint of the whole order.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-040	M	Price for an item SHALL be resolved by the following precedence, evaluated at order time: 1. Manual price override (permission-gated, audited) 2. Active promotion for	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-041	M	Price resolution SHALL be deterministic and SHALL be identical when computed offline on the device and online on the server. The resolution algorithm is part of the shared con	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-042	M	The System SHALL record which price list and which rule produced the price for each line, for audit and reporting.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-045	M	The System SHALL support discounts at line level and order level, expressed as percentage or fixed amount.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.

FR-P0S-046	M	Every discount SHALL require selection of a reason from a configurable list.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-047	M	The System SHALL enforce configurable approval thresholds:	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-048	M	Approval SHALL be obtainable by manager PIN entry on the terminal, by manager card swipe, or by remote approval request to the manager's mobile app, without abandoning the ord	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-049	M	The System SHALL record for every discount: amount, percentage, reason, applying employee, approving employee, timestamp, and order context.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-050	S	The System SHALL support "comp" (complimentary) as distinct from discount: a comped item is given free, the revenue is zero, but the cost is still recognised and inventory is	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-051	M	Discount stacking SHALL be controlled by configuration: promotions may be marked exclusive, and the System SHALL apply the most favourable single discount when exclusivity is	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-055	S	The System SHALL support automatic service charge as a percentage, configurable per branch, per order type, and conditional on guest count.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-056	S	The System SHALL support tip entry at payment, on the payment terminal, and post-payment adjustment where the acquirer supports it.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-057	S	The System SHALL support tip pooling configuration and SHALL produce a tip distribution report per shift.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-058	M	The System SHALL correctly apply the country pack's rule on whether service charge is itself taxable, as this differs by jurisdiction.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-060	M	The System SHALL support the following tender types: cash, card (via integrated terminal), card (manual/external terminal), digital wallet, gift card, loyalty points, store cr	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-061	M	The System SHALL support split payment across multiple tenders on a single order, with running balance display.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-062	M	The System SHALL support splitting a bill by: equal division among N parties, by seat, by selected items, and by arbitrary amount.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-063	M	For cash payment, the System SHALL compute change due and SHALL apply the country pack's cash rounding rule, recording any rounding adjustment as a separate ledger amount.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-064	M	Integrated card payment SHALL be initiated from the POS, and the System SHALL handle terminal timeout, decline, partial approval, and communication failure without leaving the	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-065	M	Payment operations SHALL be idempotent. A retried payment request bearing the same idempotency key SHALL NOT create a second charge. Risk: The most damaging POS defect class i	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-066	M	The System SHALL NOT store, log, or transmit card PANs, CVV, or magnetic stripe data. Only the last 4 digits, card scheme, authorisation code, and terminal reference SHALL be	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-067	S	The System SHALL support on-account sales to registered credit customers with a configurable credit limit, blocking sales that would exceed it without override permission. 8.5	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-070	M	The System SHALL distinguish four correction operations:	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-071	M	Post-fire voids SHALL prompt the user to classify the disposition of the produced item: returned to stock (if applicable), wasted, or given to staff. The classification SHALL	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P0S-072	M	Refunds SHALL reference the original order and SHALL NOT permit a refund exceeding the original amount, in aggregate across all refunds against that order.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.

FR-P05-073	M	Refunds SHALL require a reason and, above a configurable threshold, manager approval.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-074	M	Refunds SHALL be returned to the original tender type by default; refunding to a different tender SHALL require elevated permission and SHALL be flagged in the fraud detection	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-075	M	All voids, cancellations, and refunds SHALL generate audit entries containing the actor, approver, reason, amount, and full before/after state.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-080	S	The System SHALL provide a graphical floor plan editor allowing definition of areas, tables, shapes, capacities, and positions.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-081	S	The floor plan SHALL display live table state: available, seated, ordered, food served, bill requested, payment in progress, needs cleaning.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-082	S	The System SHALL support merging tables, splitting a table's order, and transferring an order between tables, each with an audit record.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-083	S	The System SHALL display time-since-seated and time-since-last-course per table, to support service pacing.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-084	C	The System SHALL support server section assignment, restricting or highlighting tables per assigned server.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-090	M	A cashier SHALL be required to open a shift, declaring an opening float, before processing sales.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-091	M	The System SHALL support mid-shift cash operations: pay-in (adding cash to drawer), pay-out (removing cash for an expense), and safe drop (removing excess cash to the safe), e	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-092	M	Safe drops SHALL be enforceable by a configurable drawer limit that triggers a prompt or a block when exceeded.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-093	M	The System SHALL produce an X report (non-resetting mid-shift summary) on demand for authorised users.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-094	M	Shift close SHALL require a physical cash count. The System SHALL support both blind count (expected amount hidden until after entry) and open count, configurable per branch.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-095	M	Blind count SHALL be the default configuration.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-096	M	Shift close SHALL compute and record cash variance, and SHALL require a reason and manager acknowledgement when variance exceeds a configurable tolerance.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-097	M	Denomination-level counting SHALL be supported, with the System computing the total from denomination counts.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-100	M	The System SHALL print a customer receipt containing all elements mandated by the branch's country pack, including tax registration number, invoice sequence, tax breakdown, an	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-101	M	Receipt layout SHALL be template-driven per country pack and per brand, supporting logo, custom header/footer text, and language selection.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-102	M	The System SHALL support bilingual receipts with configurable layout: Arabic only, English only, or both with defined ordering.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-103	M	The System SHALL support digital receipt delivery by SMS, WhatsApp, email, and QR code, in addition to or instead of printing.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-104	S	The System SHALL support receipt reprint, with reprints clearly marked as duplicates and logged.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
FR-P05-105	M	Kitchen tickets SHALL print in the kitchen's configured language, which may differ from the customer receipt language.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.

FR-POS-106	M	The System SHALL handle printer failure gracefully: queue the job, alert the user, permit continuation of the sale, and retry automatically, never blocking the transaction.	NOT IMPL.	Missing: Sales/POS domain does not exist: no order, line, payment, tender, discount, refund or void model, service or route.
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Offline & Sync — 36 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-OFF-001	M	The POS SHALL detect its operating mode continuously and SHALL indicate it unobtrusively but unambiguously in the interface.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-002	M	Mode transitions SHALL NOT interrupt an in-progress order.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-003	M	The POS SHALL operate in Isolated mode for a minimum of 72 hours without functional degradation of sales capture.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-010	M	The local database SHALL be encrypted at rest with a key derived from device credentials and a server-issued key, unusable if the device registration is revoked.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-011	M	Downward reference data SHALL be versioned, and the client SHALL request only deltas since its last known version.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-012	M	The client SHALL periodically perform a full reference-data reconciliation (checksum comparison per entity type) to detect and repair silent drift.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-013	M	Local storage SHALL be bounded: completed and fully-synced orders older than a configurable period (default 30 days) SHALL be pruned locally while remaining available from the	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-015	M	All entities created on a device SHALL receive a client-generated ULID as their permanent primary key. The server SHALL NOT reassign identifiers.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-016	M	Human-readable order numbers SHALL be allocated from server-issued blocks per terminal per business day, as specified in FR-POS-002.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-017	M	Where a jurisdiction requires a strictly gapless fiscal document sequence, the country pack SHALL specify the sequence strategy:	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-018	M	Unused numbers in an expired pre-allocated block SHALL be reported as void to the fiscal authority where required, and SHALL never be silently discarded.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-020	M	Sync SHALL be batched, with batch size bounded by plan limits and by payload size.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-021	M	Every operation SHALL carry an idempotency key. The server SHALL persist processed keys for at least 30 days and SHALL return the original result for a repeated key rather tha	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-022	M	Operations within a batch SHALL be applied in causal order. An operation whose causal parent has not been applied SHALL be deferred, not rejected.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-023	M	The server SHALL respond per-operation with accepted, duplicate, conflict, or rejected, including a reason for the latter two. A single failing operation SHALL NOT fail the ba	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-024	M	The client SHALL remove an operation from the outbox only upon receiving a definitive server response for it.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-025	M	Sync SHALL be resumable: a connection lost mid-batch SHALL resume without duplication, guaranteed by batch and operation idempotency.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-026	M	The client SHALL apply exponential backoff with jitter on failure, and SHALL prioritise financially significant operations (orders, payments, cash sessions) ahead of others in	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-030	M	The client SHALL pull reference data changes since its cursor on a configurable interval and immediately on regaining connectivity.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.

FR-OFF-031	S	Where a persistent connection is available, the server SHALL push change notifications by WebSocket so that price changes, 86'd items, and permission revocations propagate wit	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-032	M	Permission and terminal revocations SHALL be applied by the client immediately on receipt and SHALL be re-verified on every reconnection. 21.6 LAN Peer Coordination	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-035	M	POS and KDS devices on the same local network SHALL discover each other by mDNS and SHALL exchange order and bump events directly, so that the kitchen functions with no intern	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-036	M	One device per branch SHALL be electable as the LAN coordinator, holding the authoritative branch-local state during an outage. Election SHALL be automatic with deterministic	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-037	M	LAN peer communication SHALL be authenticated using a branch-scoped key distributed during registration, and SHALL be encrypted.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-038	S	Where the LAN coordinator fails, remaining devices SHALL elect a new coordinator within 30 seconds and SHALL reconcile divergent state on election.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-040	M	The System SHALL classify every entity type by conflict strategy:	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-041	M	Ordering across devices SHALL use Hybrid Logical Clocks, which preserve causality without depending on synchronised wall clocks. On local event: $l' = \max(l, \text{physical_time})$ c'	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-042	M	The System SHALL detect device clock skew exceeding a configurable threshold (default 5 minutes) on every sync, SHALL record it, SHALL alert the branch manager, and SHALL pres	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-043	M	Conflicts that cannot be resolved automatically SHALL be recorded in a conflict register, SHALL raise an alert, and SHALL be presented to a manager for manual resolution with	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-044	M	All automatic conflict resolutions SHALL be recorded in the audit log with both input states and the applied rule.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-045	M	On receiving an operation, the server SHALL revalidate all financially significant computations — prices, discounts, taxes, totals, loyalty accrual — rather than trusting clie	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-046	M	Where server revalidation produces a different result, the server SHALL accept the transaction (the sale physically occurred), record both values, and raise a reconciliation e	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-047	M	Systematic revalidation mismatches from one terminal SHALL be treated as a signal of stale reference data or client tampering, and SHALL escalate to a platform alert after a c	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-050	M	Business logic that must produce identical results on client and server SHALL be specified as a language-neutral test corpus, executed by both the Dart client test suite and t	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-051	M	Any divergence between client and server on the corpus SHALL block release.	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.
FR-OFF-052	M	Every production discrepancy detected under FR-OFF-046 SHALL be triaged, and where it reveals a logic divergence, a new corpus case SHALL be added before the fix is merged. 21	NOT IMPL.	Missing: Offline/Sync domain does not exist: no local store, sync endpoint, HLC or conflict register.

Procurement — 33 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-PRC-001	M	The System SHALL implement the procure-to-pay cycle above, with configurable steps that may be skipped per tenant policy.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.

FR-PRC-002	M	A tenant SHALL be able to operate in “simple mode” where a purchase order and goods receipt are combined into a single receiving action, for small operations that do not raise	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-005	M	The System SHALL maintain a supplier master with: code, legal name, trading name, tax registration number, addresses, contacts, payment terms, currency, delivery lead time, mi	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-006	M	The System SHALL maintain supplier price lists per item, with unit, pack size, price, validity window, and optional volume break tiers.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-007	M	The System SHALL support multiple suppliers per item with a preference ranking, and SHALL display comparative pricing when ordering.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-008	S	The System SHALL track price history per supplier per item and SHALL alert when a received price differs from the agreed price by more than a configurable tolerance.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-009	S	The System SHALL maintain a supplier scorecard computing:	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-010	S	The System SHALL support approved-supplier lists per item category, and SHALL warn or block when ordering a category from a non-approved supplier.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-011	S	The System SHALL store supplier compliance documents (food safety certificates, licences, insurance) with expiry dates and expiry alerting.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-015	S	Branches SHALL be able to raise purchase requisitions for central review, supporting the common group structure where branches request and head office buys.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-016	S	The System SHALL support consolidating requisitions from multiple branches into a single supplier purchase order, retaining branch attribution for delivery and cost allocation	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-017	M	Purchase orders SHALL contain: supplier, delivery location, expected delivery date, lines with item, quantity, unit, unit price, tax, and total.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-018	M	The System SHALL support a configurable approval workflow for purchase orders based on total value:	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-019	M	The approval workflow SHALL enforce segregation of duties: the requester SHALL NOT be an approver of their own requisition or order.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-020	M	Approval SHALL be actionable from the mobile app and from an email link with a signed, single-use, time-limited token.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-021	S	Approved purchase orders SHALL be transmittable to suppliers by email (PDF plus structured attachment), WhatsApp, or supplier portal where integrated.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-022	S	The System SHALL generate suggested purchase orders from reorder-point analysis (FR-INV-067), grouped by preferred supplier, presented for review and adjustment before approva	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-023	M	Purchase orders SHALL be amendable before receipt, with amendment history retained and re-approval required if the value increases beyond the approved band.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-030	M	The System SHALL support receiving against a purchase order, receiving without a purchase order (direct receipt), and receiving a partial quantity.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-031	M	At receipt the System SHALL capture per line: quantity received, unit, batch number, production date, expiry date, unit price if differing from PO, and rejection quantity with	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-032	M	Receipt SHALL create stock movements of type purchase_receipt and, for batch-tracked items, SHALL create batch records.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.

FR-PRC-033	M	The System SHALL enforce configurable over-receipt tolerance, requiring approval to receive more than the ordered quantity beyond the tolerance.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-034	S	The System SHALL support receiving on a mobile device with barcode scanning and camera capture of delivery notes.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-035	S	The System SHALL support temperature recording at receipt for chilled and frozen items, with configurable acceptable ranges and automatic rejection prompts outside them.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-036	M	Rejected goods SHALL create a rejection record with reason and SHALL NOT enter stock, and SHALL generate a supplier credit expectation.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-037	S	The System SHALL support supplier returns after receipt, creating a negative stock movement and a credit note expectation.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-040	M	The System SHALL support recording supplier invoices against goods receipts.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-041	M	The System SHALL perform a three-way match across purchase order, goods receipt, and supplier invoice, evaluating:	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-042	M	Invoices matching within tolerance SHALL be eligible for payment approval. Invoices outside tolerance SHALL enter a dispute state requiring resolution.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-043	S	The System SHALL support supplier credit notes, applied against outstanding invoices.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-044	S	The System SHALL maintain a supplier account statement showing invoices, credits, payments, and outstanding balance, with an ageing breakdown (current, 30, 60, 90+ days).	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-045	S	The System SHALL generate a payment proposal listing invoices due within a horizon, respecting payment terms and available early-settlement discounts.	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.
FR-PRC-046	C	The System SHALL support invoice capture by photograph with OCR extraction of supplier, date, invoice number, line items, and totals, presented for human verification before p	NOT IMPL.	Missing: Procurement domain does not exist: no supplier, purchase order, goods receipt or invoice model.

Finance & Treasury — 32 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
BR-FIN-001		Percentage computations (tax, discount, service charge) SHALL be performed at full decimal precision and rounded exactly once, at the point specified by the country pack.	NOT IMPL.	Missing: No money/rounding/tax engine
BR-FIN-002		The default rounding mode is HALF_UP (round half away from zero), overridable per country pack.	NOT IMPL.	Missing: No money/rounding/tax engine
BR-FIN-003		Where a discount is applied to an order and must be distributed across lines for tax purposes, distribution SHALL use Money.allocate so that the sum of distributed amounts exa	NOT IMPL.	Missing: No money/rounding/tax engine
BR-FIN-004		Cash rounding (to the nearest 5 or 25 minor units where the jurisdiction has withdrawn small denominations) SHALL be applied only to the cash portion of a payment and SHALL be	NOT IMPL.	Missing: No money/rounding/tax engine
BR-FIN-005		The client and server SHALL produce byte-identical monetary results for the same inputs. This is verified by the shared conformance suite in 21.9 and any divergence is a relea	NOT IMPL.	Missing: No money/rounding/tax engine
FR-FIN-001	M	The System SHALL model cash drawers as branch-level entities, with one open cash session per drawer at any time.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-002	M	A cash session SHALL be bound to exactly one employee. Two employees SHALL NOT share an open session on the same drawer.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.

FR-FIN-003	S	The System SHALL support shared-drawer mode where a branch policy permits multiple cashiers on one drawer, in which case accountability attaches to the shift rather than the i	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-004	M	Expected cash SHALL be computed as: Expected Cash = Opening Float + Cash Sales + Cash Tips (if placed in drawer) + Pay-ins – Cash Refunds – Pay-outs – Safe Drops ± Cash Roundi	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-005	M	Cash variance SHALL be computed as Counted Cash – Expected Cash and SHALL be recorded on the session.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-006	M	Variance beyond a configurable tolerance SHALL require a reason and approval by a user with cash.variance.approve, who SHALL NOT be the session owner.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-007	M	Cash sessions SHALL be immutable once closed. Corrections SHALL be recorded as adjusting entries referencing the session.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-010	M	The System SHALL record, per session and per day, totals by tender type: cash, each card scheme, each wallet, gift card, voucher, on-account, and aggregator-settled.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-011	S	The System SHALL support reconciliation of card totals against the payment terminal's batch settlement report, flagging differences.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-012	S	The System SHALL support reconciliation of aggregator-settled orders against aggregator payout statements, computing expected payout net of commission and flagging discrepanci	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-015	S	The System SHALL support recording branch operating expenses with category, amount, payment method, supplier, date, attachment, and approval status.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-016	S	Petty cash paid from the drawer SHALL be recorded as a pay-out linked to the expense record, so that the drawer reconciles.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-017	S	Expenses above a configurable threshold SHALL require approval before posting.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-018	S	The System SHALL support recurring expense templates for rent, utilities, and subscriptions.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-020	M	The System SHALL support a business-day close operation per branch.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-021	M	Day close SHALL be blocked while any cash session remains open, and SHALL list the blocking sessions.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-022	M	Day close SHALL produce a Z report containing: gross sales, discounts, refunds, net sales, tax by rate, sales by category, sales by tender, sales by order type, transaction co	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-023	M	Z reports SHALL be sequentially numbered per branch, immutable, and retrievable for any historical date.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-024	M	The System SHALL support a configurable business-day boundary per branch (e.g. 04:00), so that late-night trading is attributed to the correct operating day.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-025	S	Day close SHALL be performable automatically at the configured boundary where the branch enables it, with any open sessions force-closed and flagged.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-026	M	Day close SHALL trigger: fiscal document finalisation, inventory day-end snapshot, report pre-aggregation, and accounting export generation where configured.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-030	M	Tax SHALL be computed by the country pack's tax engine, never by hard- coded logic.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-031	M	The System SHALL support tax-inclusive and tax-exclusive pricing, configurable per branch and per price list.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.

FR-FIN-032	M	The System SHALL support multiple simultaneous tax components (e.g. VAT plus municipality fee plus service tax), each with its own rate, base, and rounding.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-033	M	The System SHALL support tax classes per item (standard, reduced, zero- rated, exempt) and SHALL support order-type-dependent rates where the jurisdiction differentiates dine-	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-034	M	Tax SHALL be computed at line level and summed, not computed on the order total.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.
FR-FIN-035	M	The rounding mode and rounding point SHALL be specified by the country pack and SHALL be applied consistently across POS, server, receipt, and fiscal submission.	NOT IMPL.	Missing: Finance/Treasury domain does not exist: no cash session, till, expense or Z-report model.

Costing & Analytics – 31 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-CST-001	M	On order completion, the System SHALL compute COGS by expanding each order line's recipe to base ingredients, applying modifier recipe deltas, and valuing consumption at the i	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-002	M	COGS SHALL be recorded on the order line as unit_cost_snapshot and SHALL NOT be recomputed retroactively when ingredient costs change.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-003	M	The System SHALL report food cost percentage as: $\text{Food Cost \%} = \text{COGS} \div \text{Net Sales} \times 100$	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-004	M	Food cost percentage SHALL be reportable by item, category, branch, brand, day-part, order type, and date range.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-005	M	The System SHALL compute contribution margin per item: $\text{Contribution Margin} = \text{Selling Price (ex-tax)} - \text{Direct Cost Contribution Margin \%} = \text{Contr}$	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-006	S	The System SHALL support inclusion of configurable non-recipe direct costs in the margin calculation: packaging (per order type), delivery commission (per channel), and dispos	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-007	S	The System SHALL compute channel profitability, showing revenue, COGS, commission, packaging, and net contribution per sales channel.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-010	M	For a defined period and location, the System SHALL compute theoretical usage per stock item as the sum, across all sold items, of recipe-expanded ingredient quantities.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-011	M	For the same period and location, the System SHALL compute actual usage as: $\text{Actual Usage} = \text{Opening Count} + \text{Purchases Received} + \text{Transfers In} + \text{Production Output} - \text{Transfers Ou}$	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-012	M	The System SHALL compute variance: $\text{Variance Quantity} = \text{Actual Usage} - \text{Theoretical Usage}$ $\text{Variance Value} = \text{Variance Quantity} \times \text{Current Unit Cost}$ $\text{Variance Percentage} = \text{Var}$	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-013	M	Recorded waste SHALL be presented as a separate column so that the report distinguishes explained from unexplained variance: $\text{Unexplained Variance} = \text{Variance Quantity} - \text{Recorded}$	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-014	M	The variance report SHALL be sortable by variance value descending, so that the largest financial losses appear first regardless of quantity.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-015	S	The System SHALL classify variance by likely cause using the following heuristics, presented as hypotheses rather than conclusions:	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-016	S	The System SHALL compute variance at a per-shift granularity where opening and closing counts are available at that granularity, and SHALL fall back to the count period otherw	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-017	S	The System SHALL provide a variance trend view showing an item's variance percentage over successive count periods. 13.4 Waste Analysis	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.

FR-CST-020	M	The System SHALL report waste by value and quantity, grouped by item, reason, category, employee, station, day-part, shift, and branch.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-021	M	The System SHALL compute waste as a percentage of both COGS and net sales.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-022	S	The System SHALL compute a rolling baseline of expected waste per item and location, and SHALL flag deviations beyond a configurable number of standard deviations.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-023	S	The System SHALL produce a "cost of waste" summary translating waste into the revenue required to offset it: Revenue Required to Offset = Waste Value ÷ (Average Contribution M	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-024	C	The System SHALL correlate waste by reason with operational factors (staffing level, order volume, new employee presence, equipment fault reports) to support root- cause analy	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-030	S	The System SHALL compute labour cost per shift from attendance records and configured hourly rates.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-031	S	The System SHALL compute:	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-032	S	The System SHALL display sales and labour on a common hourly axis so over- and under-staffing by hour is visible.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-035	S	The System SHALL produce a branch profitability summary: Gross Sales – Discounts – Refunds = Net Sales (excl. tax) – COGS = Gross Profit – Labour Cost = Contribution After Lab	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-036	S	The System SHALL support recording branch operating expenses (rent, utilities, maintenance) with category, recurrence, and allocation.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-037	S	The System SHALL support branch ranking by any metric, with configurable normalisation (absolute, per seat, per square metre, per labour hour) so that branches of different si	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-038	C	The System SHALL compute a break-even sales level per branch from fixed costs and average contribution margin, and SHALL show progress against it during the month.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-040	S	The System SHALL monitor and flag the following patterns for review. Every flag is a prompt for human investigation, never an accusation.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-041	M	Anomaly flags SHALL be visible only to users holding an explicit governance.view_anomalies permission.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-042	M	The System SHALL present anomaly flags with the underlying evidence — the specific transactions, timestamps, and comparison baseline — so that a manager can evaluate them rath	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.
FR-CST-043	S	Baselines SHALL be computed per branch and per role, not globally, because normal void rates differ enormously between a fast-food counter and a fine-dining room.	NOT IMPL.	Missing: Costing/Analytics domain does not exist; also blocked on Sales and Production Spec.

Workforce — 29 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-HRM-001	M	The System SHALL maintain employee records containing: employee code, names (localised), national/residency identifier, contact details, emergency contact, date of birth, hire	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-002	M	The System SHALL support employment types: full-time, part-time, casual, contractor, and trainee, with different rules for scheduling and overtime.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-003	M	The System SHALL store compensation basis (hourly rate, monthly salary, or per-shift rate) with effective dating, and SHALL restrict visibility of compensation to holders of a	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.

FR-HRM-004	S	The System SHALL store document references with expiry dates (residency permit, work permit, health certificate, food-handling licence) and SHALL alert before expiry.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-005	M	Employees SHALL be assignable to multiple branches with a designated home branch, supporting the common practice of staff covering shifts across a group.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-006	M	Employee records SHALL be deactivatable but SHALL NOT be deletable while historical transactions reference them.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-010	S	The System SHALL support creating shift schedules by branch, week, position, and employee.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-011	S	The System SHALL support shift templates and week-pattern copying to avoid rebuilding a schedule each week.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-012	S	The System SHALL validate schedules against configurable rules and warn or block on violation:	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-013	S	The System SHALL project scheduled labour cost for the week and compare it against a configurable target percentage of forecast sales.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-014	S	The System SHALL forecast required staffing per hour from historical sales patterns for the same weekday and season, and SHALL overlay this against the drafted schedule.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-015	S	The System SHALL publish schedules to employees by mobile notification and SHALL record acknowledgement.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-016	C	The System SHALL support shift swap requests between employees, subject to rule validation and manager approval.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-017	S	The System SHALL support leave requests, approval workflow, and a leave balance per configured leave type.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-020	M	The System SHALL support clock-in and clock-out via: POS terminal with PIN, mobile app with geofence validation, and biometric device integration where available.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-021	M	The System SHALL record clock events with timestamp, method, terminal or device, and — where mobile — GPS coordinates.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-022	M	The System SHALL detect and flag: late arrival beyond a grace period, early departure, missing clock-out, clock-in outside the geofence, and clock-in with no scheduled shift.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-023	M	The System SHALL prevent a clock-in more than a configurable interval before the scheduled shift start, to prevent inflated paid hours.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-024	S	Missing clock-outs SHALL be auto-closed at a configurable maximum shift length, flagged for manager correction, and SHALL NOT silently accrue hours.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-025	M	Manual attendance corrections SHALL require a reason, SHALL be permission-gated, and SHALL create an audit entry retaining the original value.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-026	S	The System SHALL support break tracking, distinguishing paid from unpaid breaks per configuration.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-027	C	The System SHALL support photo capture at clock-in as a deterrent to buddy-punching, subject to explicit configuration and employee notice.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-030	S	The System SHALL compute per-employee metrics: total sales, order count, average order value, items per order, upsell attachment rate, average service time, void count and val	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-031	S	The System SHALL compute kitchen employee metrics from KDS data: items prepared, average preparation time by item, and remake count.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.

FR-HRM-032	S	The System SHALL support ranking employees on any metric within a branch, position, and period.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-033	M	The System SHALL compute worked hours, split into regular and overtime per the branch's country pack overtime rules.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-034	M	Overtime beyond a configurable threshold SHALL require pre-approval, and unapproved overtime SHALL be flagged in a separate report rather than silently included.	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-035	M	The System SHALL export payroll data in configurable formats (CSV with mappable columns, plus provider-specific templates) containing employee identifier, period, regular hour	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.
FR-HRM-036	S	The System SHALL NOT compute net pay, tax withholding, or statutory contributions. It exports inputs to payroll; it is not a payroll engine. Chapter 15 — Roles, Permissions &	NOT IMPL.	Missing: Workforce domain does not exist: no employee, shift, roster or payroll model.

CRM & Loyalty — 26 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-CRM-001	M	The System SHALL maintain customer records with: name, phone (primary identifier), email, addresses, date of birth, preferred language, tags, and consent flags.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-002	M	Phone number SHALL be the primary identifier, unique within a tenant, stored in E.164 format.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-003	M	Customer creation from the POS SHALL require no more than phone number and name, completable in under 15 seconds.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-004	M	The System SHALL maintain per-customer order history with full detail, retrievable from the POS.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-005	S	The System SHALL compute per-customer: total spend, order count, average order value, first and last order dates, favourite items, preferred branch, preferred order type, and	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-006	S	The System SHALL support multiple delivery addresses per customer with labels and delivery notes.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-007	S	The System SHALL support customer blocking, preventing on-account sales or delivery to a flagged customer, with reason and audit record.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-008	M	The System SHALL record explicit consent flags per communication channel and per purpose, with timestamp and source, and SHALL NOT send marketing communication absent recorded	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-009	M	The System SHALL support customer data export and erasure requests, with erasure anonymising the customer record while retaining transactional and fiscal records as legally re	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-015	S	The System SHALL support a points-based loyalty programme with configurable earn rate, redemption rate, expiry, and eligibility rules.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-016	S	Points SHALL be earnable on net spend excluding tax and excluding already-discounted amounts, configurable per tenant.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-017	S	Points SHALL be redeemable as a tender type at the POS, converting to a monetary discount at the configured rate.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-018	S	The System SHALL support tiered loyalty with configurable thresholds, tier benefits, and tier review periods.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-019	S	The System SHALL support a stamp-card model ("buy 9 coffees, get the 10th free") as an alternative to points, configurable per item or item group.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-020	M	Loyalty balances SHALL be maintained as an append-only transaction ledger, not as a mutable balance field.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.

FR-CRM-021	M	Offline loyalty accrual SHALL be permitted. Offline redemption SHALL be permitted up to a configurable per-transaction limit, with the risk of overdraw accepted and reported o	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-022	S	The System SHALL provide the customer a loyalty balance view via a QR code on the receipt leading to a lightweight web page requiring no app installation.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-025	S	The System SHALL support a promotion engine with conditions and effects. Conditions: date and time range, day of week, branch set, order type, channel, minimum order value, sp	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-026	S	Promotions SHALL support usage limits: total redemptions, per-customer redemptions, and per-day redemptions.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-027	M	Promotion evaluation SHALL be deterministic and identical offline and online, and is covered by the shared conformance suite.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-028	S	The System SHALL support single-use and multi-use coupon codes, with bulk generation and per-code tracking.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-029	S	The System SHALL report promotion performance: redemptions, incremental revenue, discount cost, margin impact, and redeeming customer profile.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-030	S	Promotion stacking SHALL be governed by explicit configuration; the default SHALL be non-stackable with best-value selection.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-035	C	The System SHALL compute RFM segmentation (Recency, Frequency, Monetary) and SHALL classify customers into actionable segments: champions, loyal, at-risk, hibernating, and new	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-036	C	The System SHALL compute a churn indicator based on a customer's own historical inter-purchase interval rather than a global threshold.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.
FR-CRM-037	C	The System SHALL support exporting a segment for a marketing campaign, subject to consent flags.	NOT IMPL.	Missing: CRM domain does not exist: no customer, loyalty or campaign model.

Kitchen Ops (KDS) — 23 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-KDS-001	M	The System SHALL support definition of preparation stations per branch, with configurable name, display colour, and capacity. Standard station types: Grill, Fryer, Cold/Salad,	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-010	M	The System SHALL route each order line to one or more stations, resolved by the following precedence: 1. Explicit line-level station override 2. Modifier-driven routin	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-011	M	A single order line SHALL be routable to multiple stations when the item requires multi-station preparation (a burger requiring grill and packaging).	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-012	S	The System SHALL support a Prep-Time-Aware coordination mode in which items on one order are released to their stations at staggered times, calculated backwards from a target	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-013	S	The System SHALL support an Expediter (Pass) display showing complete orders, with per-station completion state, for final assembly and quality check.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-020	M	The KDS SHALL display tickets as cards containing: order number, order type, elapsed time, table or customer reference, item lines with quantity and modifiers, and preparation	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-021	M	Modifiers SHALL be visually distinguished from item names, with removals (- no onion) rendered differently from additions (+ extra cheese).	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-022	M	Tickets SHALL be colour-coded by elapsed time against a configurable target: within target (neutral), approaching (amber), exceeded (red), critically exceeded (red, flashing).	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.

FR-KDS-023	M	The System SHALL support the following ticket sort orders, configurable per station: oldest first (FIFO), by target completion time, by order type priority, and by course sequ	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-024	M	Staff SHALL be able to mark an individual item ready ("bump item") or an entire ticket ready ("bump all").	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-025	M	The System SHALL provide a recall function restoring the most recently bumped tickets, retained for a configurable period (default 30 minutes).	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-026	M	Bump actions SHALL require a deliberate interaction (long-press, double- tap, or a dedicated confirm zone) to prevent accidental bumping from splashes or brushes against the s	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-027	S	The System SHALL support order priority flags (rush, VIP, remake) with distinct visual treatment.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-028	S	Amendments to a fired order SHALL appear as a visually distinct update on the existing ticket, with an audible and visual alert, never as a new ticket.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-029	M	Cancelled lines SHALL be struck through and highlighted on the station display, with an alert, and SHALL remain visible for a configurable period so the cook stops preparing.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-030	S	The System SHALL support "all-day" counts: an aggregate view showing total outstanding quantity per item across all active tickets at this station.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-031	S	The KDS SHALL support an icon-and-image mode for kitchens where staff literacy in the configured language cannot be assumed.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-040	M	The System SHALL record the following timestamps per ticket and per line: created, routed, first viewed, started, ready, bumped, and served.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-041	M	The System SHALL compute and report: average preparation time by item, by station, by hour, by employee, and by order type.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-042	M	The System SHALL define and report "ticket time" as bump time minus fire time, and "order time" as last-line-ready minus order-open.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-043	S	The System SHALL identify the bottleneck station per hour, defined as the station with the highest ratio of queue depth to throughput.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-044	S	The System SHALL support configurable per-item target preparation times, defaulting to the recipe's prep_time_seconds.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.
FR-KDS-045	C	The System SHALL provide a capacity warning when queued items at a station exceed configured throughput capacity for the next 15 minutes.	NOT IMPL.	Missing: Kitchen Ops domain does not exist: no ticket, bump, station display or routing runtime.

Reporting — 18 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-RPT-001	M	Analytical queries SHALL execute against a read replica, never the transactional primary.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-002	M	The System SHALL maintain pre-aggregated rollups at hourly, daily, weekly, and monthly grain for the core fact tables.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-003	M	Rollups SHALL be incrementally updated as new transactions arrive, and SHALL be fully rebuildable from source on demand.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-004	M	Every report SHALL display the timestamp of the data it reflects, and SHALL indicate when data is not yet complete for the period shown.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-005	M	Dimension tables SHALL be slowly-changing (Type 2) for attributes that affect historical interpretation — item category, employee position, branch region — so that reclassify	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.

FR-RPT-030	M	The System SHALL provide role-appropriate default dashboards.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-031	S	The Executive dashboard SHALL present: net sales against target, prime cost %, branch ranking, top and bottom performing items, exception count, and trend sparklines.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-032	S	The Branch Manager dashboard SHALL present: today's sales against forecast, hourly sales curve with labour overlay, food cost trend, items requiring reorder, expiry watch, ope	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-033	S	The System SHALL provide a live operations view: open orders, table states, kitchen queue depth, average current wait, and active terminals.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-034	C	Dashboards SHALL be customisable by the user through widget selection and arrangement, with a tenant-level default layout per role.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-040	S	The System SHALL support scheduled report delivery by email and by mobile push, with configurable recipients, schedule, and format.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-041	S	The System SHALL support a daily digest ("morning brief") delivered at a configurable time, summarising the prior business day in a form readable on a phone in under 30 second	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-042	M	Every aggregate figure SHALL support drill-down to the contributing transactions in no more than four interactions.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-043	M	The System SHALL support export in CSV, XLSX, and PDF, with exports of more than 50,000 rows processed asynchronously and delivered by notification.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-044	M	All exports SHALL be logged in the audit trail with the requesting user, filters applied, and row count.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-045	S	The System SHALL support configurable alerts delivered in real time:	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-046	M	Alerts SHALL be rate-limited and de-duplicated per recipient per type per period.	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.
FR-RPT-047	C	The System SHALL support a natural-language query interface over the reporting layer, translating questions such as "which branch had the worst food cost last week" into param	NOT IMPL.	Missing: Reporting domain does not exist: no report engine, scheduling or export.

DevOps & Release — 15 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-OPS-001	M	Deployments SHALL be zero-downtime and SHALL be rollback-capable within 5 minutes.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-002	M	Canary deployments SHALL automatically roll back on error-rate or latency SLO breach without human intervention.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-003	M	All infrastructure SHALL be defined as code (Terraform), reviewed, and version controlled. Manual production changes are prohibited except under a documented break-glass proce	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-004	M	Container images SHALL be signed, scanned, and accompanied by a Software Bill of Materials.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-005	M	Feature flags SHALL gate risky changes, permitting per-tenant progressive rollout independent of deployment.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-010	M	POS and KDS client releases SHALL support staged rollout by tenant, branch, and terminal.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-011	M	The client SHALL be able to operate against the previous API version for at least one release cycle, so that a client update failure never prevents trading.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-012	M	The System SHALL support forced client updates for security-critical releases, with the update deferrable until after the current shift closes.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.

FR-OPS-013	M	Release notes SHALL be published per release, in Arabic and English, and SHALL be visible in the product.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-020	M	The System SHALL maintain an on-call rotation with documented escalation and runbooks for every paging alert.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-021	M	Incidents SHALL be classified by severity with defined response times:	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-022	M	Every SEV-1 and SEV-2 SHALL produce a blameless post-incident review within 5 working days, with action items tracked to completion.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-023	M	A public status page SHALL report current and historical availability.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-030	S	Infrastructure cost SHALL be attributable per tenant, enabling gross- margin measurement per customer and detection of unprofitable accounts.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.
FR-OPS-031	S	The System SHALL alert when a single tenant's infrastructure consumption exceeds a configured multiple of their subscription revenue.	NOT IMPL.	Missing: No CI/CD, IaC, feature flags, release or incident tooling in the repository.

NFR - Usability — 11 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
NFR-USA-001	M	A three-line dine-in order with no modifiers SHALL be completable in no more than 6 user interactions from an empty screen.	NOT IMPL.	Missing: Client/UI requirement; no client surface exists in this backend repository.
NFR-USA-002	M	All primary touch targets SHALL be at least 48 x 48 dp with at least 8 dp separation.	NOT IMPL.	Missing: Client/UI requirement; no client surface exists in this backend repository.
NFR-USA-003		Signup to first order ≤ 30 minutes median	NOT IMPL.	Missing: Client/UI requirement; no client surface exists in this backend repository.
NFR-USA-004	M	The POS SHALL be fully operable by a user who cannot read, using a configurable image-based menu grid.	NOT IMPL.	Missing: Client/UI requirement; no client surface exists in this backend repository.
NFR-USA-005		New cashier productive after ≤ 30 minutes of training	NOT IMPL.	Missing: Client/UI requirement; no client surface exists in this backend repository.
NFR-USA-006	M	All KDS text conveying item identity SHALL be legible at 2 metres on a 21- inch display at 1920x1080, implying a minimum effective font size of 24 pt for item names.	NOT IMPL.	Missing: Client/UI requirement; no client surface exists in this backend repository.
NFR-USA-007		Dashboard meets WCAG 2.1 Level AA	NOT IMPL.	Missing: Client/UI requirement; no client surface exists in this backend repository.
NFR-USA-008		Contrast ratio ≥ 4.5:1 for text, ≥ 3:1 for UI components	NOT IMPL.	Missing: Client/UI requirement; no client surface exists in this backend repository.
NFR-USA-009		Dashboard fully keyboard-navigable	NOT IMPL.	Missing: Client/UI requirement; no client surface exists in this backend repository.
NFR-USA-010		Every destructive action is confirmable or undoable	NOT IMPL.	Missing: Client/UI requirement; no client surface exists in this backend repository.
NFR-USA-011		Every error message states what happened and what to do next 27.5 Maintainability ID Requirement Target NFR-MAINT-001	NOT IMPL.	Missing: Client/UI requirement; no client surface exists in this backend repository.

NFR - Reliability — 7 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
NFR-REL-001	M	The POS SHALL survive abrupt power loss with no loss of committed orders. Every state transition SHALL be persisted to durable local storage before the UI acknowledges it.	NOT IMPL.	Missing: No durability, idempotency or failover mechanism implemented.
NFR-REL-002	M	The KDS SHALL continue to display and accept bump actions for already- received tickets during a network outage, buffering bump events locally.	NOT IMPL.	Missing: No durability, idempotency or failover mechanism implemented.
NFR-REL-003	M	Where the KDS and POS are on the same local network, order routing SHALL function via local peer discovery even when the internet connection is down.	NOT IMPL.	Missing: No durability, idempotency or failover mechanism implemented.
NFR-REL-010	M	No committed sale SHALL be lost under any single-device failure, including abrupt power loss, application crash, or storage exhaustion.	NOT IMPL.	Missing: No durability, idempotency or failover mechanism implemented.

NFR-REL-011	M	The System SHALL guarantee at-most-once financial effect for any operation, enforced by idempotency keys.	NOT IMPL.	Missing: No durability, idempotency or failover mechanism implemented.
NFR-REL-012		Data durability ≥ 99.999999999% (managed storage guarantee)	NOT IMPL.	Missing: No durability, idempotency or failover mechanism implemented.
NFR-REL-013		RPO / RTO ≤ 5 min / ≤ 60 min	NOT IMPL.	Missing: No durability, idempotency or failover mechanism implemented.

Integrations — 6 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-INT-001	M	Every external system SHALL be accessed through an Anti-Corruption Layer that translates between the external model and the ROS domain model. External data structures SHALL NO	NOT IMPL.	Missing: No integration surface: no aggregator, accounting, payment or webhook connectors.
FR-INT-002	M	All outbound integration calls SHALL be dispatched through the transactional outbox and SHALL be idempotent.	NOT IMPL.	Missing: No integration surface: no aggregator, accounting, payment or webhook connectors.
FR-INT-003	M	All inbound integration calls SHALL be authenticated, rate-limited, validated against a schema, and idempotent by an external reference key.	NOT IMPL.	Missing: No integration surface: no aggregator, accounting, payment or webhook connectors.
FR-INT-004	M	Integration failures SHALL be visible in an integration health dashboard showing per-connector status, last success, error rate, and queue depth.	NOT IMPL.	Missing: No integration surface: no aggregator, accounting, payment or webhook connectors.
FR-INT-005	M	Integration credentials SHALL be stored encrypted, SHALL be rotatable without downtime, and SHALL never appear in logs or error messages.	NOT IMPL.	Missing: No integration surface: no aggregator, accounting, payment or webhook connectors.
FR-INT-006	M	Each connector SHALL implement a circuit breaker: after a configured failure threshold, calls are suspended and queued, with periodic probing for recovery.	NOT IMPL.	Missing: No integration surface: no aggregator, accounting, payment or webhook connectors.

NFR - Portability — 6 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
NFR-PORT-001		POS supported on Android 9+, iOS 15+, Windows 10+	NOT IMPL.	Missing: Client platform/deployment portability requirement; no client or deployment automation.
NFR-PORT-002		KDS supported on Android 9+ and evergreen browsers ID Requirement	NOT IMPL.	Missing: Client platform/deployment portability requirement; no client or deployment automation.
NFR-PORT-003		Dashboard supported on the two most recent versions of Chrome, Safari, Edge, and Firefox	NOT IMPL.	Missing: Client platform/deployment portability requirement; no client or deployment automation.
NFR-PORT-004		Dashboard responsive from 360 px to 2560 px	NOT IMPL.	Missing: Client platform/deployment portability requirement; no client or deployment automation.
NFR-PORT-005		Deployment portable across major cloud providers (no proprietary lock-in beyond managed Postgres, Redis, object storage)	NOT IMPL.	Missing: Client platform/deployment portability requirement; no client or deployment automation.
NFR-PORT-006		ESC/POS printer support across the certified hardware matrix	NOT IMPL.	Missing: Client platform/deployment portability requirement; no client or deployment automation.

Quality Assurance — 5 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
FR-QA-001	M	The domain layer SHALL be tested exclusively with unit tests requiring no database, no HTTP, and no framework.	NOT IMPL.	Missing: No QA automation beyond unit/E2E suites; requirements describe process and coverage gates that do not exist.
FR-QA-002	M	Every functional requirement SHALL trace to at least one automated test, verified by the traceability matrix in Appendix B.	NOT IMPL.	Missing: No QA automation beyond unit/E2E suites; requirements describe process and coverage gates that do not exist.
FR-QA-010	M	The System SHALL provide reproducible seed datasets: a minimal single-branch café, a 12-branch chain, a multi-brand multi-country group, and a cloud kitchen with aggregators.	NOT IMPL.	Missing: No QA automation beyond unit/E2E suites; requirements describe process and coverage gates that do not exist.
FR-QA-011	M	Test data SHALL be synthetic. Production data SHALL NOT be copied into non-production environments.	NOT IMPL.	Missing: No QA automation beyond unit/E2E suites; requirements describe process and coverage gates that do not exist.

FR-QA-012	M	Where production-shaped data is required for performance testing, it SHALL be generated at production scale rather than copied, or anonymised through a documented, verified pi	NOT IMPL.	Missing: No QA automation beyond unit/E2E suites; requirements describe process and coverage gates that do not exist.
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NFR - Capacity — 1 requirements · 0% implemented

ID	Pri	Requirement (SRS wording, abridged)	Status	Evidence / missing pieces / dependencies
NFR-CAP-001	M	The local store SHALL accommodate at least 20,000 orders and their lines without performance degradation beyond stated budgets.	NOT IMPL.	Missing: Local store requirement; no client exists.